

Discrete Response, Time Series and Panel Data - Notes

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W-01-Timeseries-Introduction-Visualisation-Stationarity

 Note

Coming soon...

W-02-Timeseries-Introduction-Visualisation-Stationarity

Log-transformations

Instead of the original time series X_t , consider the log-transformed time series with the elements.

$$g_t = \log(X_t)$$

When to apply it?:

- When the distribution of the time series elements is right-skewed.
- When a time series could be produced by the product of model outcomes.

Differencing

Let's take a linear function (defined on a set of integer numbers)

$$X_t = m \cdot t + b$$

Then, the backward difference

$$X_t - X_{t-1} = (m \cdot t + b) - (m \cdot (t-1) + b) = m$$

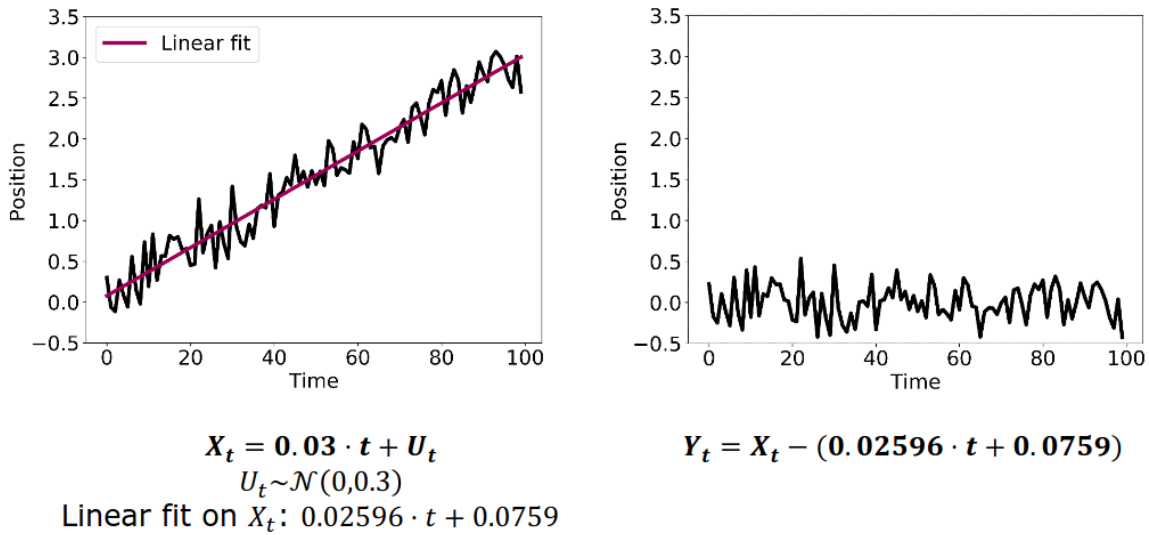
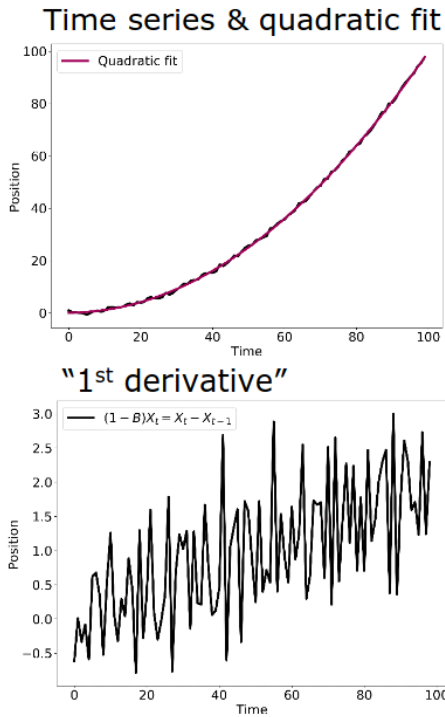


Figure 1: Differencing

Higher Order Differencing

Use multiple derivatives to eliminate lower-order terms. Trend order is reduced, but the stationary part may be more complex. After trend is eliminated, maybe a new dependencies in the stationary part.

Example



Initial time series: $X_t = 0.01 t^2 + U_t$ with $U_t \sim \mathcal{U}(-1,1)$

"1st derivative" displays still linear dependence, where as "2nd derivative" is "constant."

Note that a residual process does not have structure of U_t .

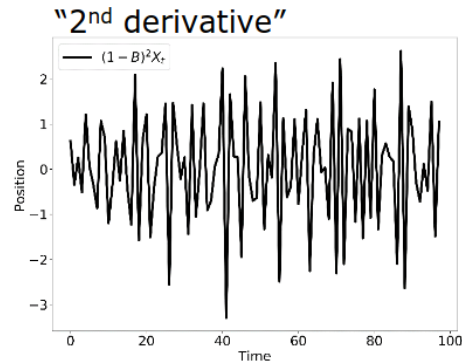
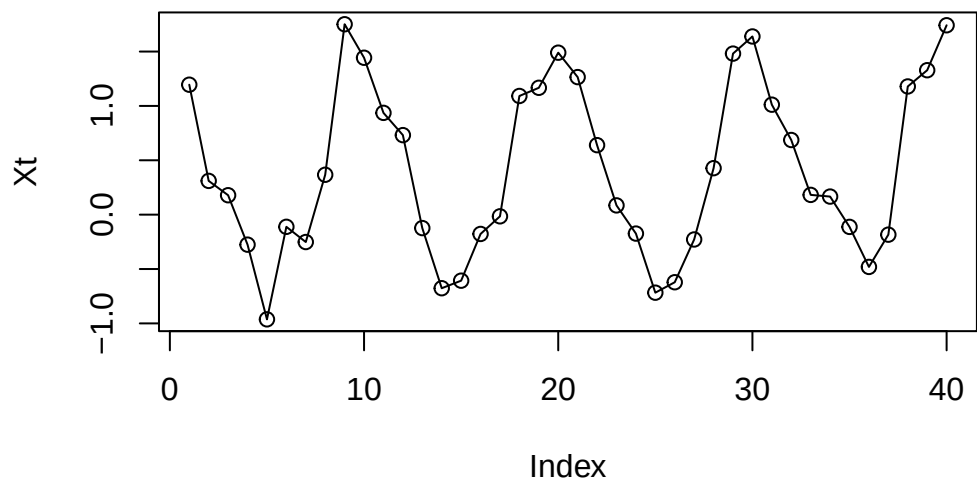


Figure 2: Remove Higher Order Trend

Differencing Seasonality

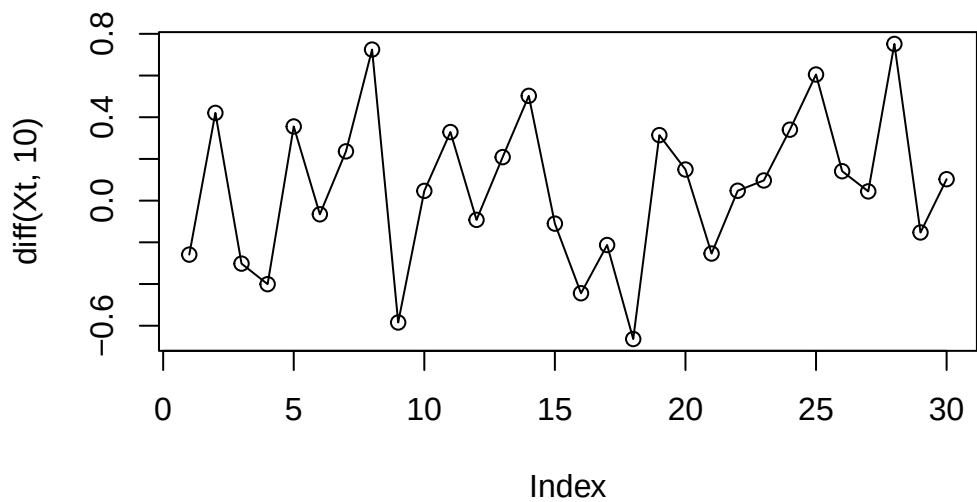
Seasonality manifests itself by a repetitive behavior after a fixed lag time p .

```
tRange<-1:40
Xt<-cos(2*pi/10*tRange)+runif(40)
plot(Xt,type="o")
```



What happens if we differentiate now with lag 10?

```
plot(diff(Xt,10),type="o")
```



Summary of differencing

Advantages:

- can remove trends and seasonality effects
- is easy to use

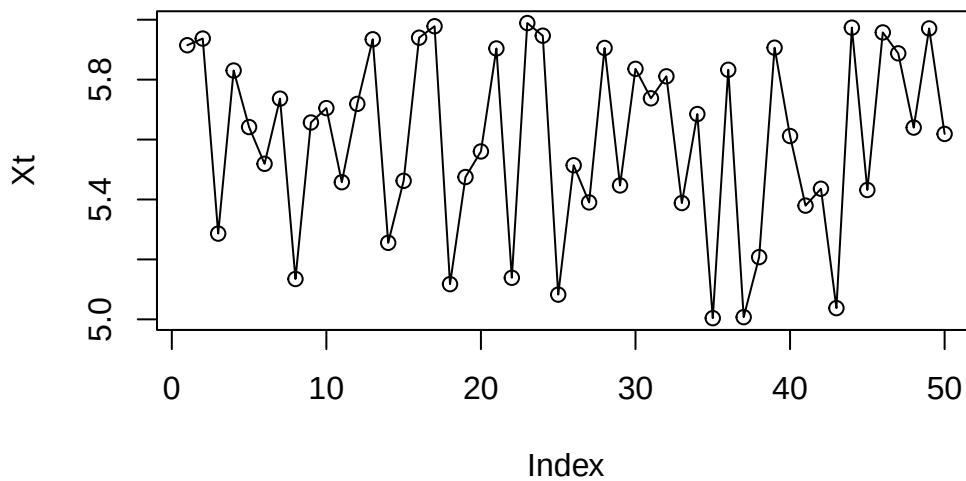
Disadvantages:

- the resulting time series is shorter than the original one
- the results of differencing may display dependencies among different elements
- the original decomposition elements are unknown

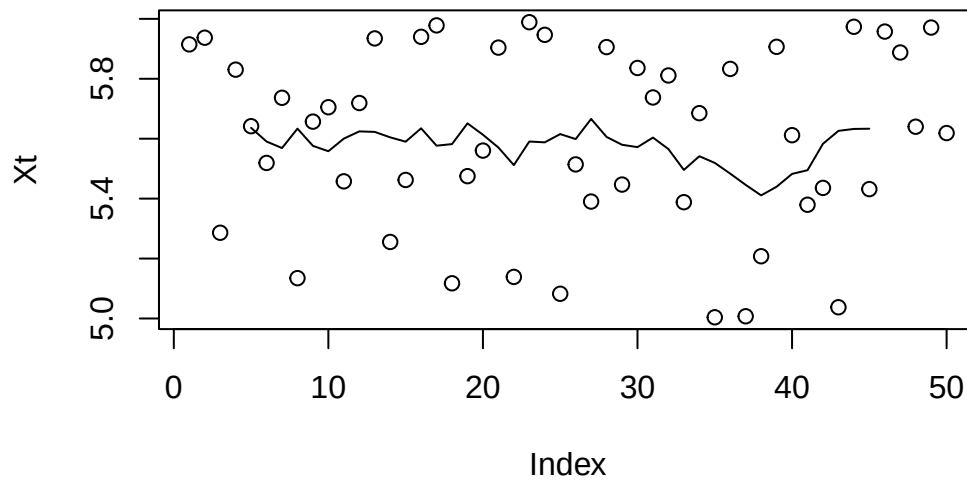
Filtration

To reduce the noise element in an experiment, you can repeat the measurement multiple times and then calculate the average.

```
set.seed(42)
Xt<-5+runif(50)
plot(Xt,type="o")
```



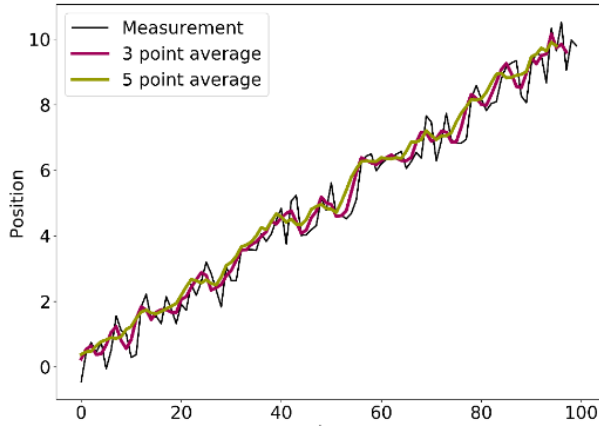
```
filtered<-filter(Xt,filter=rep(1,10)/10)
plot(Xt)
points(filtered,type="l")
```



Reasons for using Filtration

By averaging measurements, we obtain a new averaged time series. By increasing the number of averaged measurements, the parts converges to the expectation value of the measurement error, i.e. 0.

Note: The short-term variations are smoothed out and the trend is confirmed.



Initial time series:

$$X_t = 0.1 t + U_t, U_t \sim \mathcal{U}(-1,1)$$

$$\text{3 point average: } a_0 = a_{-1} = a_{-2} = \frac{1}{3}$$

$$\text{5 point average: } a_0 = a_{-1} = a_{-2} = \frac{1}{5} \\ = a_{-3} = a_{-4} = \frac{1}{5}$$

Figure 3: Filtering

Linear filters to quantify seasonality effects

In case of seasonality effects, the filter has to expand over the full period to extract the trend. To quantify the seasonal impact, the values of the same time in the season have to be averaged.

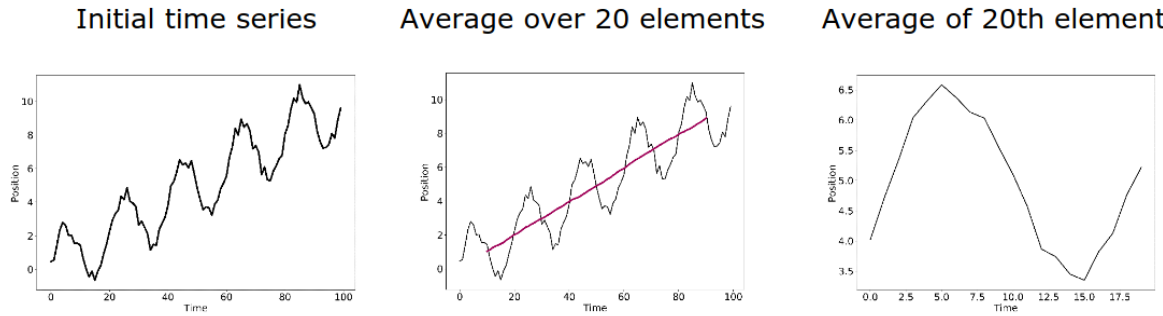


Figure 4: Quantify Seasonality Effects

Seasonal-trend decomposition by Loess Algorithm (STL)

Linear filters are great if periodicity is obvious; but in less obvious cases, extracting the individual components may be laborious. An alternative is the seasonal-trend decomposition procedure by Loess: An iterative, non-parametric smoothing algorithm that is more robust.

Fitting of Periodic Function

If you know the generative model, you can use fitting to determine the contributions of trend and seasonality. Fitting methods (among others):

- Ordinary least square (OLS)
- Robust fitting (to prevent the impact of outliers)
- GLS (generalised least square) / time series regression methods

```
set.seed(42)
#Generate the sample data set
t<-seq(1,100,length=100)
data<-0.1*t+cos(2*pi/10*t)+runif(100)
ts<-ts(data)
#Fit the model
fit<-lm(ts~t+cos(2*pi/10*t))
summary(fit)
```

Call:

```
lm(formula = ts ~ t + cos(2 * pi/10 * t))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.58266	-0.22271	0.04478	0.25411	0.49285

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.639744	0.059840	10.69	<2e-16 ***
t	0.097718	0.001029	94.98	<2e-16 ***
cos(2 * pi/10 * t)	0.973247	0.041999	23.17	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2969 on 97 degrees of freedom

Multiple R-squared: 0.9901, Adjusted R-squared: 0.9899

F-statistic: 4836 on 2 and 97 DF, p-value: < 2.2e-16

Autocorrelation

The autocorrelation depends only on the lag for weakly or strongly stationary time series.

$$p(k) := \text{Cor}(X_{t+k}, X_t)$$

Interpretation of autocorrelation

The correlation $p(k)^2$ squared corresponds to the percentage of variability explained by the linear association between X_t and X_{t+k} .

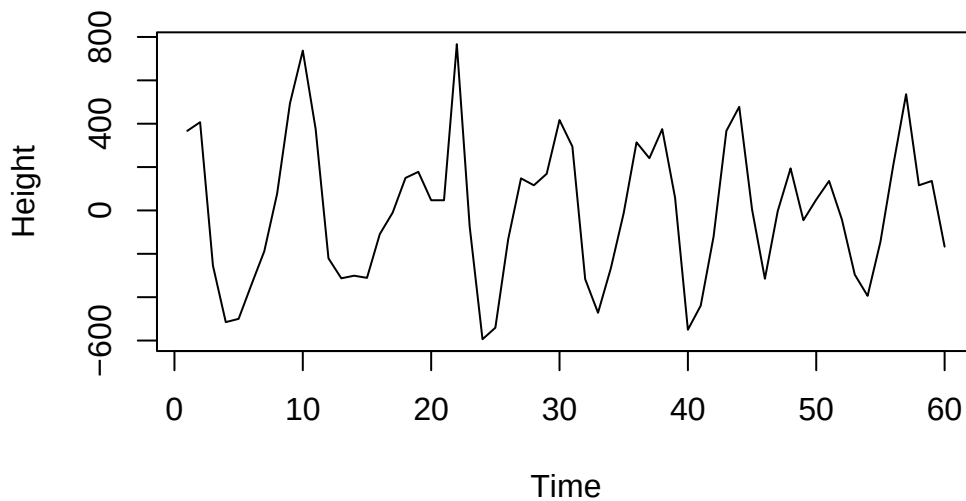
Lagged Scatter Plot

Exchange average over realizations with average over time.

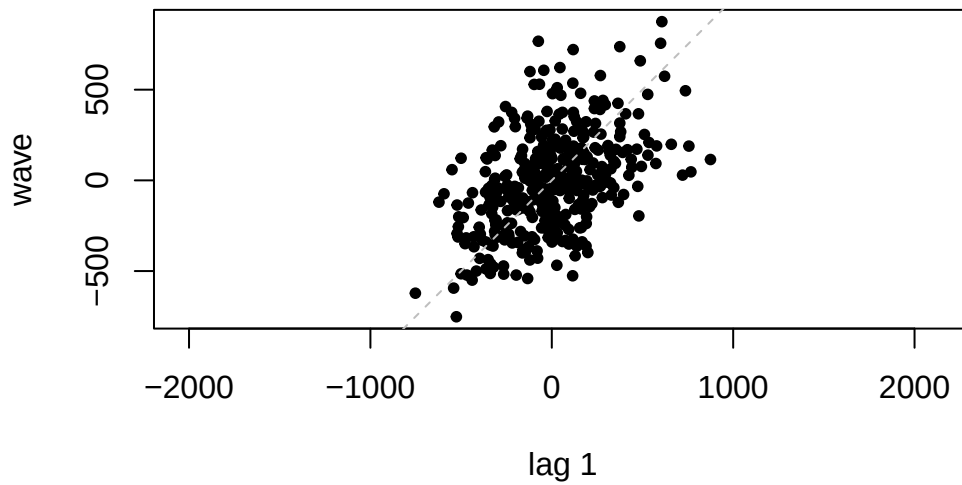
```
address <- "https://raw.githubusercontent.com/AtefOuni/ts/master/Data/wave.dat"
dat <- read.table(address,header=T)

#Generate ts object
wave<-ts(dat$waveht)

#Visualize the data
plot(window(wave,1,60),ylab="Height")
```

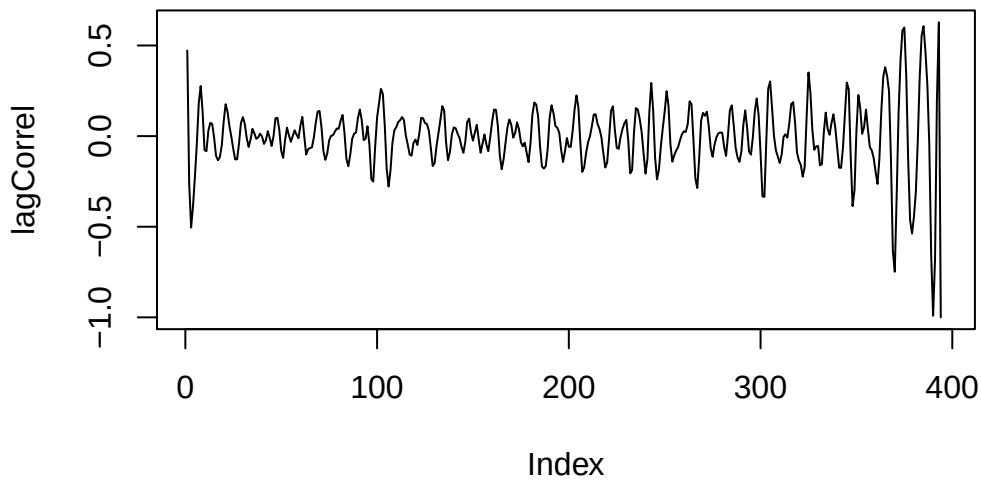


```
#Lag plot  
lag.plot(wave,do.lines=FALSE,pch=20)
```



Calculate lagged scatter plot estimator

```
# Calculate the Pearson Correlation coefficients  
n <- length(wave)  
lagCorrel <- rep(0,n)  
  
for (i in 1:(n-1)){  
  lagCorrel[i]=cor(wave[1:(n-i)],wave[(i+1):n])  
}  
  
plot(lagCorrel,type="l")
```



Plug-In Estimation

The plug-in estimator is the standard approach to computation autocorrelations. Calculated with an `acf` function in R.

Summary Autocorrelation

- (Auto)correlation measures the linear association at a given lag
- For realizations, the autocorrelation is estimated by running the calculations over time instances instead of realizations (therefore stationarity is desired)
 - Lagged scatter plot estimator
 - Plug-in estimator
- For high lags, the lagged scatter plot estimator diverges (because too few time points are considered)
- The plug-in estimator corrects the diverging behavior but generates a bias.

⚠ Warning

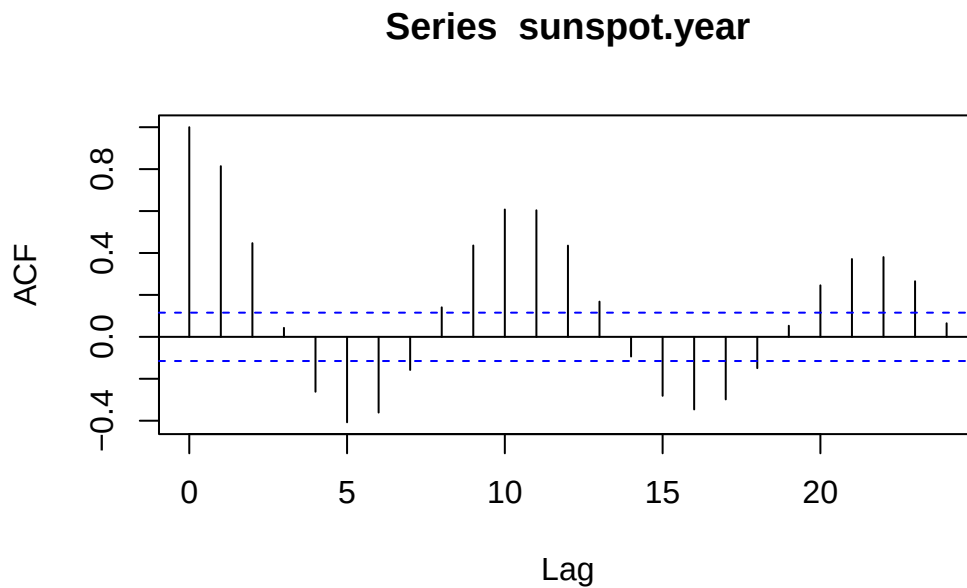
Warning: Non-linear relations may confuse the autocorrelation.

W-03-Timeseries-ACF-AR

Correlogram

Standard command: `acf(data)` for plug-in estimator values.

```
data(sunspot.year)
acf(sunspot.year)
```



Caution

The first lag at lag 0 always have to be a value of 1

Bounds of Autocorrelation

Under the null hypothesis of a time series process with iid distributed random variables, a 95 % acceptance region for the null hypothesis is:

$$\pm \frac{1.96}{\sqrt{n}}$$

For stationary processes, plug-in estimator value $\hat{p}(k)$ within the confidence band $\pm \frac{1.96}{\sqrt{n}}$ are considered to be different from 0 only by chance.

Ljung-Box Test

The Ljung-Box test assesses the null hypothesis that the random variables are independent when the h autocorrelation coefficients $\hat{p}(k)$ for $k = 1, \dots, h$ are equal to zero.

```
Box.test(sunspot.year, lag=10, type="Ljung-Box")
```

Box-Ljung test

```
data: sunspot.year  
X-squared = 542.41, df = 10, p-value < 2.2e-16
```

Characteristic Patterns in ACF

Patterns in ACF for non-stationary time series

For non-stationary time series, the ACF decays only slowly.

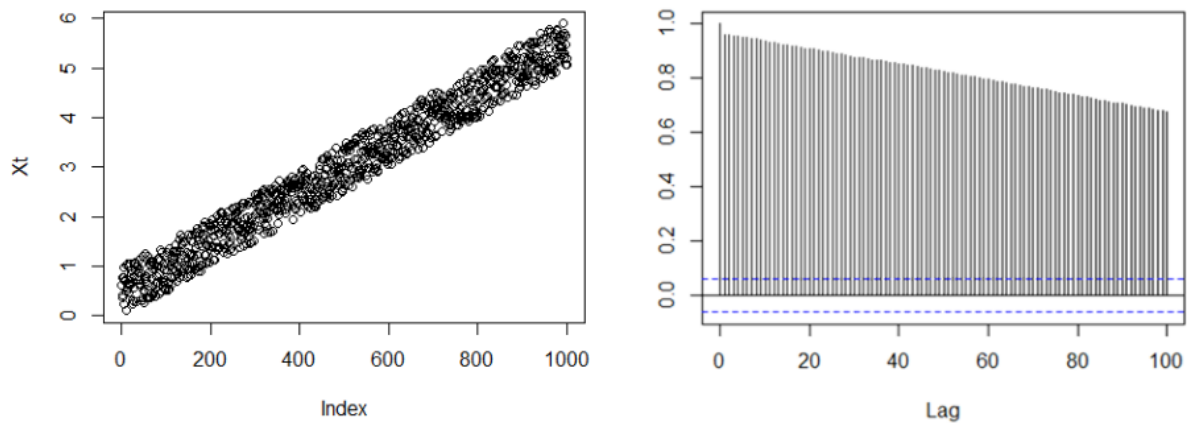


Figure 5: Non-Stationary time series

ACF of time series with seasonal pattern

For time series with seasonality, the ACF displays periodic structure.

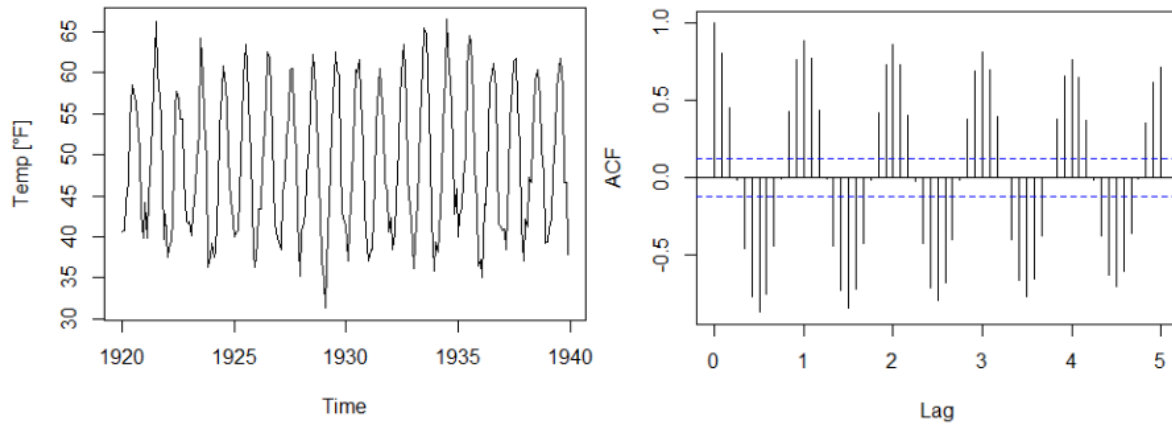


Figure 6: Time series with seasonal pattern

ACF of time series with seasonal pattern and trend

Time series with seasonality and trend, the ACF displays periodicity and slow decay.

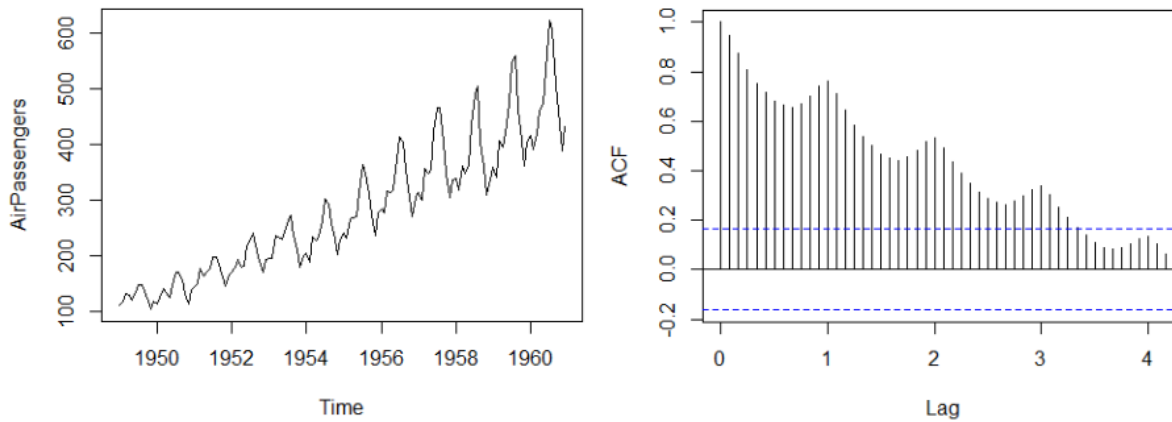


Figure 7: Time series with seasonal pattern and trend

ACF of time series with outliers

Outliers may be diagnosed by lagged scatter plots. Note that each outlier causes other outliers.

! Important

The plug-in estimator $\hat{p}(k)$ is sensitive to outliers.

Properties of ACF

Autocorrelation of process vs. estimated values of realization

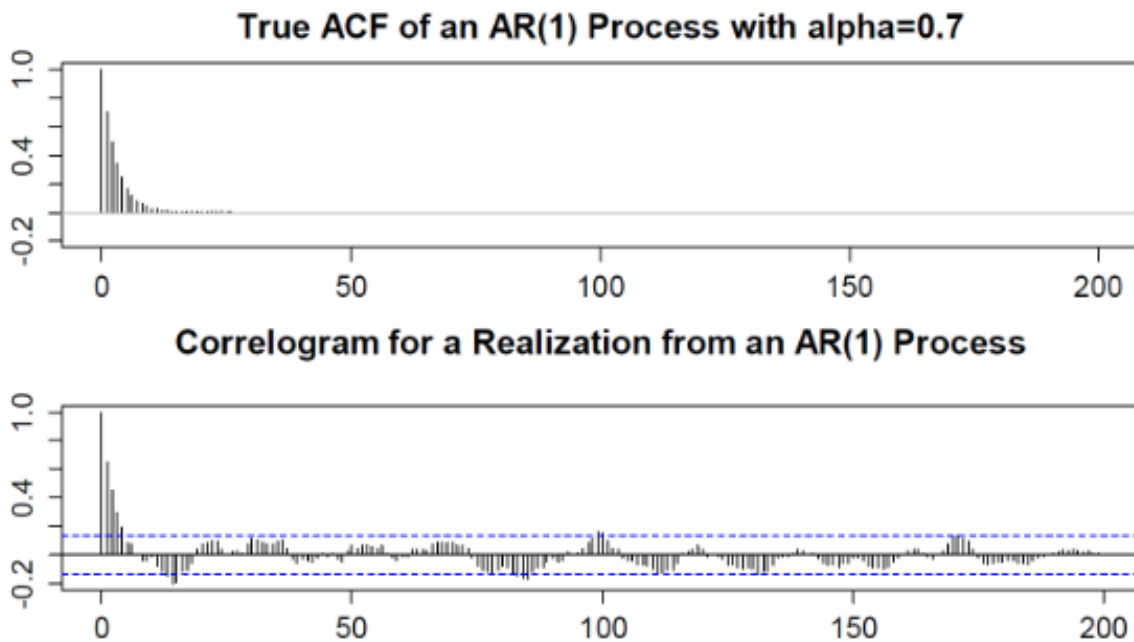


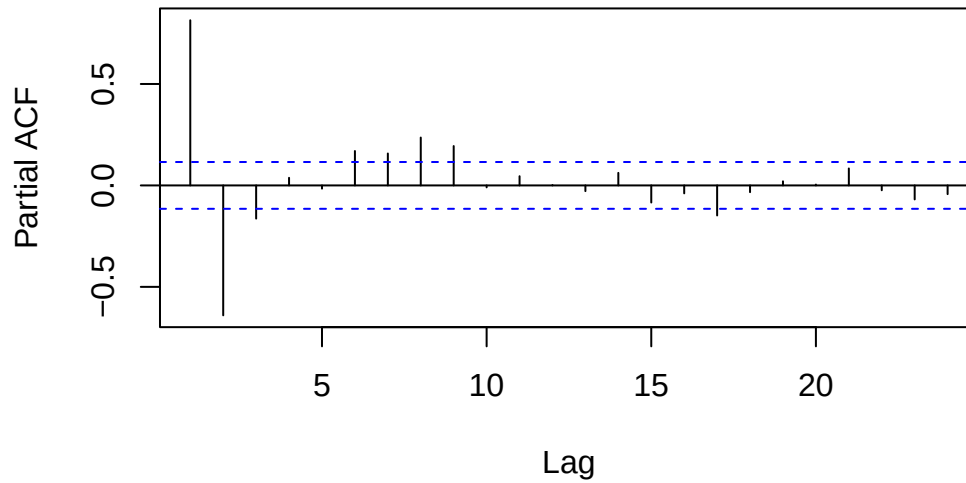
Figure 8: Correlogram of one realization compare with the true plot

Partial Autocorrelation

A challenge in the interpretation of ACFs is that the effect of smaller lags remains. The partial autocorrelation function (PACF) gives the partial correlation of a stationary time series with its own lagged values, regressed the values of the time series at all shorter lags.

```
pacf(sunspot.year)
```

Series sunspot.year



Note

- ACF -> MA-Model
- PACF -> AR-Model

Analysis of the ARMA(2,1) Process for Stationarity

Given the process equation:

$$X_t = 1.2X_{t-1} - 0.36X_{t-2} + E_t + 0.5E_{t-1}$$

Step 1: Isolate the AR Part

To check for stationarity, only the AR (autoregressive) part is relevant. We move all X terms to the left side of the equation (paying close attention to the signs):

$$X_t - 1.2X_{t-1} + 0.36X_{t-2} = E_t + 0.5E_{t-1}$$

Step 2: Set up the Characteristic Polynomial

We replace the time lags with the variable z (where $X_t \rightarrow 1$, $X_{t-1} \rightarrow z$, and $X_{t-2} \rightarrow z^2$):

$$\Phi(z) = 1 - 1.2z + 0.36z^2$$

Step 3: Calculate the Characteristic Roots

To find the roots (zeros), we set the polynomial equal to zero:

$$0.36z^2 - 1.2z + 1 = 0$$

Applying the **quadratic formula** (abc-formula) with $a = 0.36$, $b = -1.2$, and $c = 1$:

$$z = \frac{-(-1.2) \pm \sqrt{(-1.2)^2 - 4 \cdot 0.36 \cdot 1}}{2 \cdot 0.36}$$

$$z = \frac{1.2 \pm \sqrt{1.44 - 1.44}}{0.72}$$

Since the discriminant under the square root is 0, we have a **repeated root** (double root):

$$z_1 = z_2 = \frac{1.2}{0.72} = \frac{5}{3} \approx 1.67$$

Step 4: Check the Stationarity Condition

An ARMA process is stationary if the absolute value (modulus) of all characteristic roots is **greater than 1** ($|z_i| > 1$), meaning they lie outside the unit circle.

- **Condition Check:** $|1.67| > 1$ ✓

Conclusion

Since all characteristic roots lie outside the unit circle, the given ARMA process is **stationary**.

W-04-Timeseries-ACF-AR

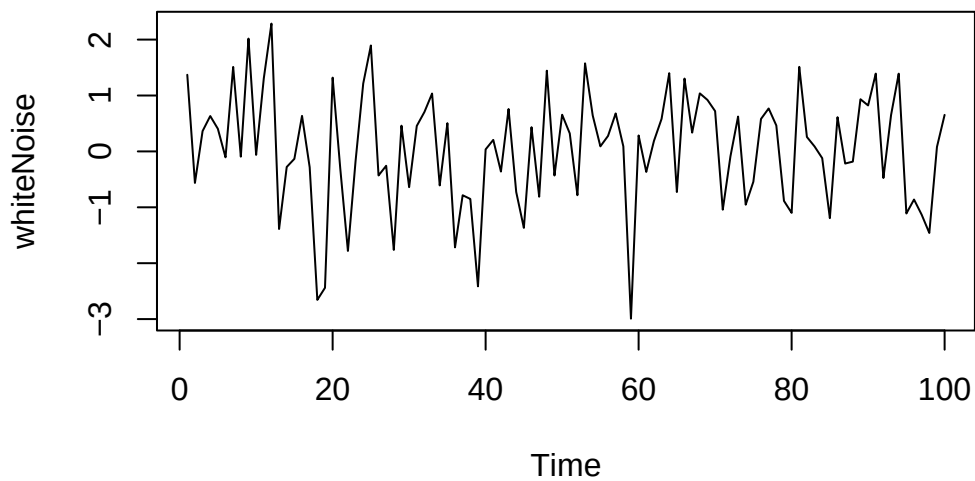
Fundamentals of Modeling

White noise

A time series process $W_t, t \in \mathbb{N}$ is called white noise if it is a sequence of independent and identically distributed random variables with mean 0.

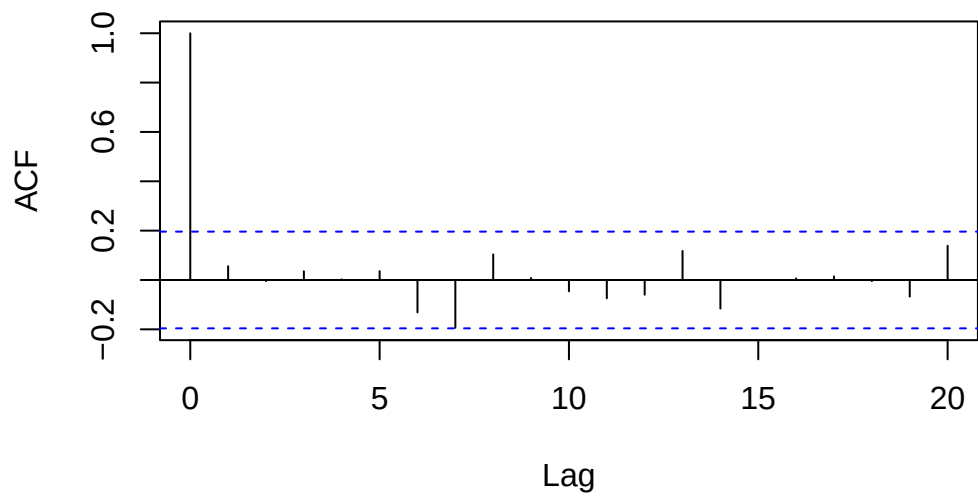
Note: The name “white” indicates that all frequencies are included, as in the case of “white light.”

```
set.seed(42)
whiteNoise <- ts(rnorm(100,mean=0,sd=1))
plot(whiteNoise)
```



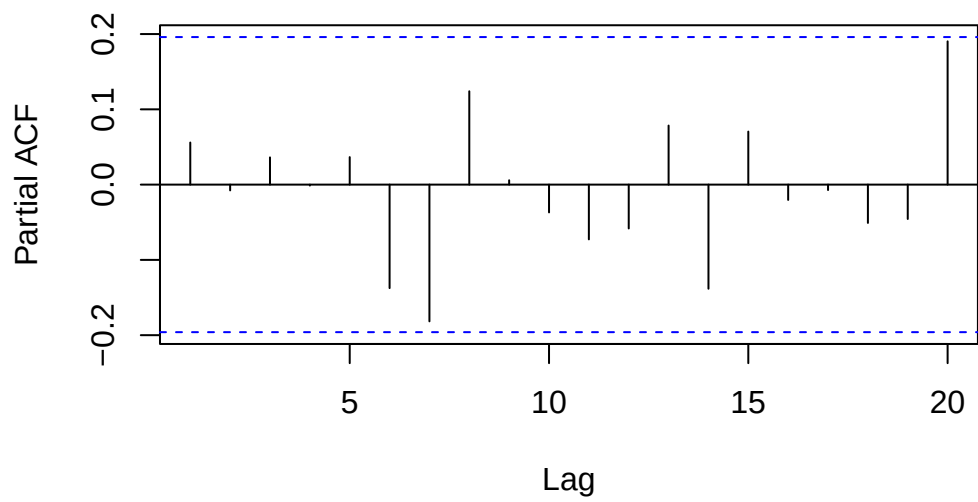
```
acf(whiteNoise)
```

Series whiteNoise



```
pacf(whiteNoise)
```

Series whiteNoise



Causal White Noise

A series of *iid* random variables e_t , which are individually independent of the proceeding time series elements X_t

Basic Time Series Models

The following models are often used when there is limited information about how time series data is generated:

- AR: auto-regressive model
- MA: moving average model
- ARMA: combination of AR and MA model
- ARIMA: non-stationary ARMA model (I=integration)
- SARIMA: seasonal ARIMA model

Auto-Regression Models (AR)

Express the current random variable as a linear combination of the last p time steps and a “completely independent” term e_t called innovation.

$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t$$

where α_x is a set of finite parameters and e_t a series of independent random variables.

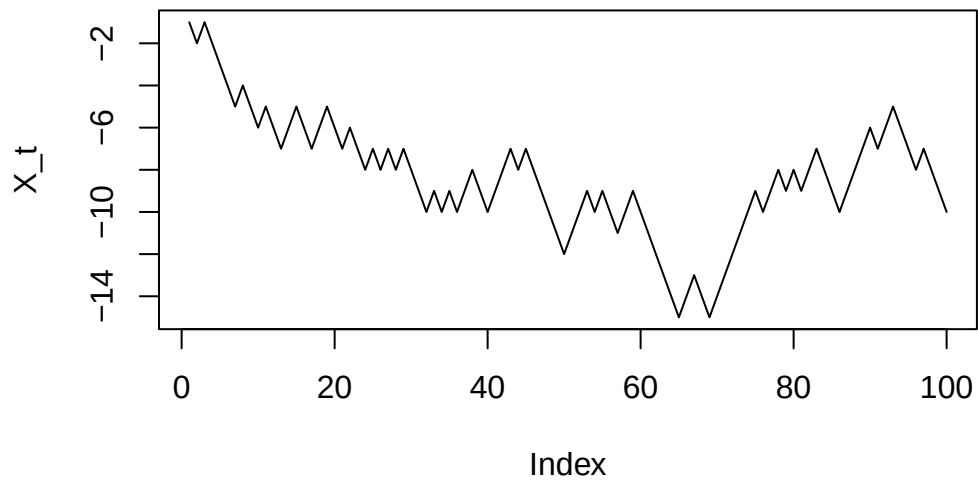
AR(1) as Random Walk Model

Trajectory of an object moving along a line. At each moment, the object takes randomly the decision in which direction to move on.

$$X_t = X_{t-1} + e_t$$

where e_t is random with equal probability for -1 and 1 .

```
set.seed(42)
step<- 1 - 2 * round(runif(100))
X_t<-cumsum(step)
plot(X_t, type="lines")
```

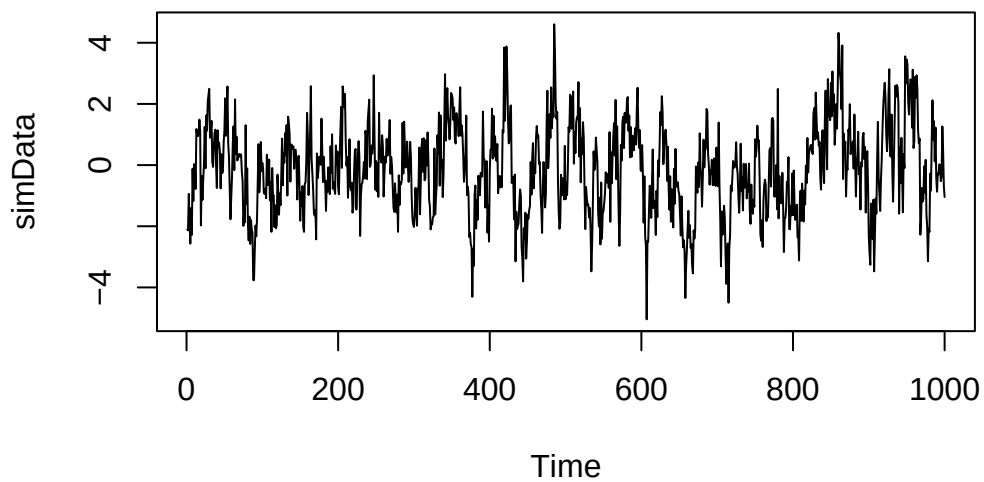


Simulation $AR(p)$ Processes

We would like to simulate:

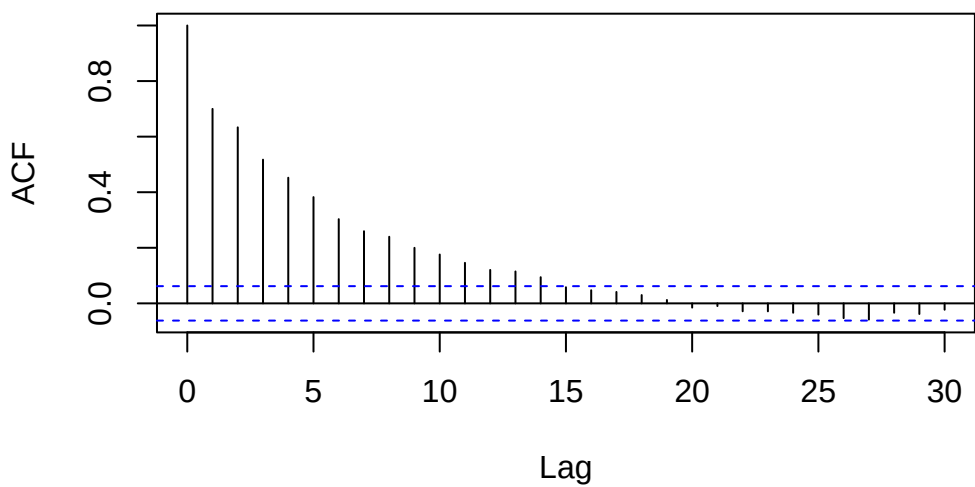
$$X_t = 0.5X_{t-1} + 0.3X_{t-2} + e_t \text{ with } e_t \sim N(0, 1) \text{ iid.}$$

```
set.seed(42)
ar.coefficients <- c(0.5, 0.3)
simData <- arima.sim(n=1000,model=list(ar=ar.coefficients))
plot(simData)
```

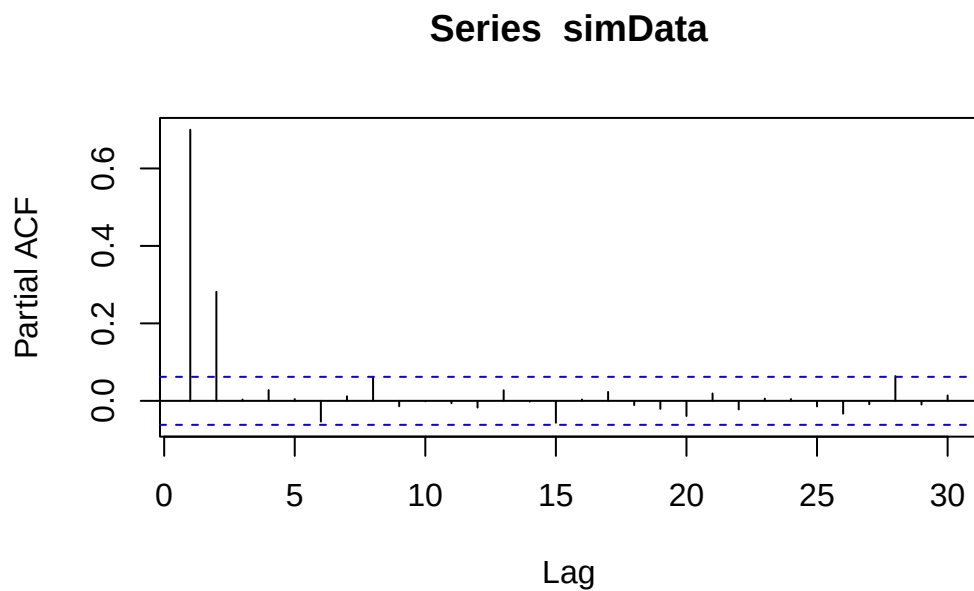


```
acf(simData)
```

Series simData



```
pacf(simData)
```



Stationarity of $AR(p)$ Models

$AR(p)$ models are stationary when the:

- mean is 0
- absolute value of the roots of the characteristic polynomial is all larger than 1

```
polyroot(c(1,-0.8,-0.4))
```

```
[1] 0.8708287+0i -2.8708287+0i
```

As the first value has an absolute value below 1, the time series is not stationary.

Key Properties of $AR(p)$ Models

The $AR(1)$ process:

$$X_t = 0.75 * X_t + e_t$$

is stationary because the characteristic polynomial:

```
polyroot(c(1,-0.75))
```

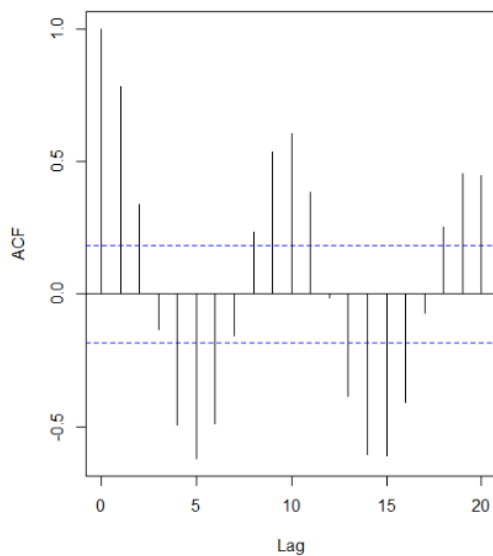
```
[1] 1.333333+0i
```

For the model identification of $AR(p)$ processes, two properties are key:

- Autocorrelation decays quickly or displays a damped sinusoid
- Partial autocorrelation function has a clear cut-off at p

Model Order Identification

Plug-in estimated ACF of $\log(\text{lynx})$



Partial ACF

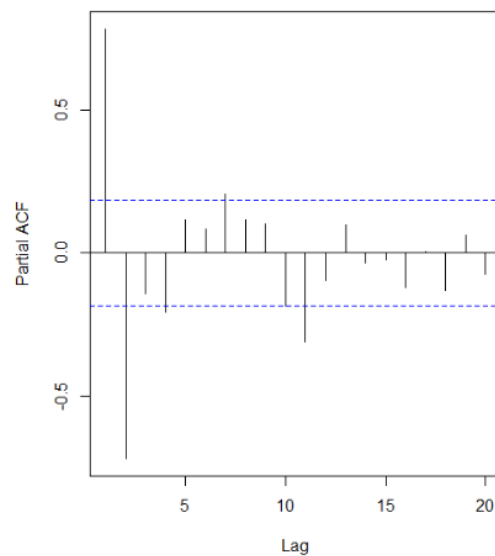


Figure 9: Plug-in estimated ACF and PACF

- ACF displays damped sinusoid
- PACF has clear cut-off after lag 11, 7, 4 or 2 (depending on the threshold).

Try fitting an $AR(2)$, $AR(4)$, $AR(7)$, $AR(11)$

W-05-Timeseries-AR-MA

Model Parameter Identification

Time series are often not centered: I.e. instead of a classical AR(p) process, a centered version has to be fitted.

$$Y_t = m + X_t = m + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t$$

Fitting parameters are: - the global mean m - the AR-coefficients $\alpha_1, \dots, \alpha_p$ - the parameters defining the distribution of the innovation e_t . Here, we assume a normal distribution.

Model Parameter from Ordinary Least Square (OLS)

We want to estimate the parameters from the model equation, we have a total of $n - p$ points for this, because we have a total of n time segments and each equation links p time instance.

The OLS procedure is:

1. Estimate the global mean $\hat{m} = \bar{y} = \frac{1}{n} \sum_{t=1}^n y_t$ and determine $x_t = y_t - \hat{m}$
2. Carry out a regression on x_t without intercept. The resulting parameters are $\hat{\alpha}_1, \dots, \hat{\alpha}_p$
3. Estimate the standard deviation of the innovation $\hat{\sigma}_E^2$ deviation of the residual.

```
ar.ols(data, order=p)
```

Yule-Walker Equations

Basic idea: The model coefficients and the ACF values are entangled. The values of the auto-correlation function depend on the model coefficients. A careful calculation for an AR(p) model yields the following equations, called Yule-Walker equations:

$$\sum_{i=1}^p \alpha_i \rho(k-i) = \rho(k) \quad \text{for } k > 0$$

$$\sum_{i=1}^p \alpha_i \rho(i) = \rho(0) - \sigma_E^2 \quad \text{for } k = 0$$

Note: Note that the autocorrelation function is symmetric, i.e. $\rho(-k) = \rho(k)$ for all k .

To fit the AR(p) model parameters, we must solve the above equations with the estimated values of the auto correlation function.

```
ar.yw(data,order=p)
```

Further options for fitting

Burg's algorithm

Offers another fitting cost function that exploit also the first p function values.

```
ar.burg(data,order.max=p)
```

Maximum likelihood estimator (MLE)

Determines the parameter based on an optimization of the maximum likelihood function

```
arima(Data,order=c(2,0,0))
```

Which method to use when?

One option is to use the Akaike information criterion (AIC), which estimates the relative amount of information lost by the considered model. The less information a model loses, the higher the quality of that model.

```
f.burg<-ar.burg(log(lynx))
plot(
  0:f.burg$order.max,
  f.burg$aic,
  type="l"
)
```



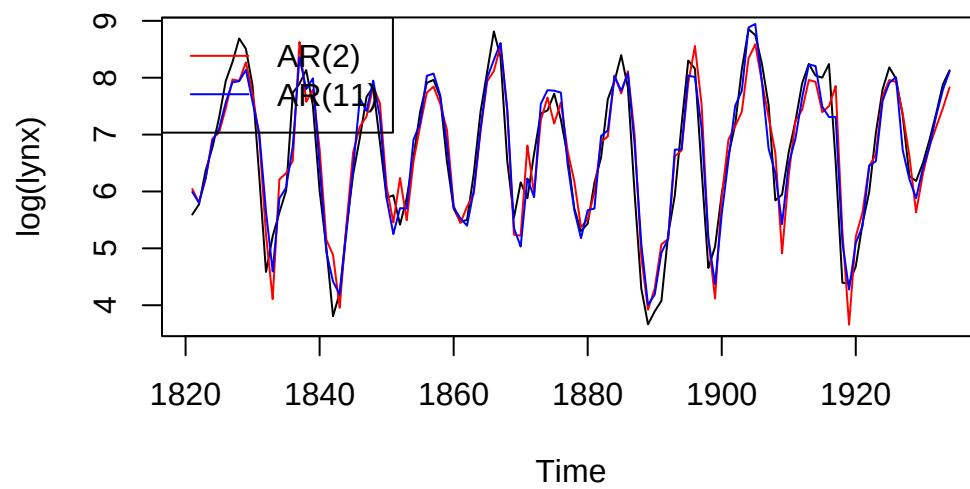
Note: No order parameter is given any more.

Model Diagnostic

Example

```
# Fitting the two models
fit.2<-arima(log(lynx),order=c(2,0,0))
fit.11<-arima(log(lynx),order=c(11,0,0))

# Display the time series and model predictions
plot(log(lynx))
lines(log(lynx) - fit.2$resid, col = "red")
lines(log(lynx) - fit.11$resid, col = "blue")
legend("topleft",
      legend = c("AR(2)", "AR(11)"),
      col = c("red", "blue"),
      lty = 1)
```



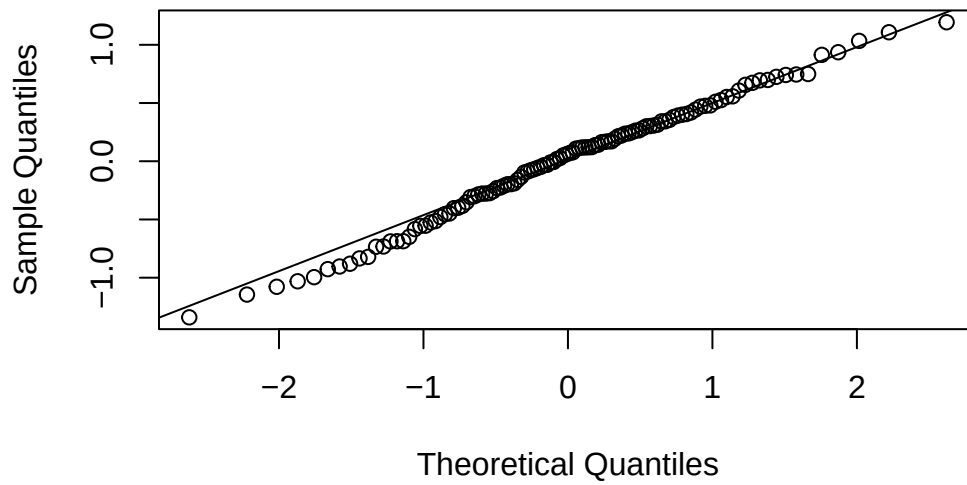
QQ-Plot

Compare quantile of measurement data with theoretical distribution

- X-Axis: Theoretical distribution
- Y-Axis: Actual data distribution

```
qqnorm(fit.2$residuals,main="Normal Q-Q Plot: AR(2)")
qqline(fit.2$residuals)
```

Normal Q-Q Plot: AR(2)



Moving Average Models (MA)

A moving average MA(q) model is a model in the form

$$X_t = E_t + \beta_1 E_{t-1} + \beta_2 E_{t-2} + \cdots + \beta_q E_{t-q}$$

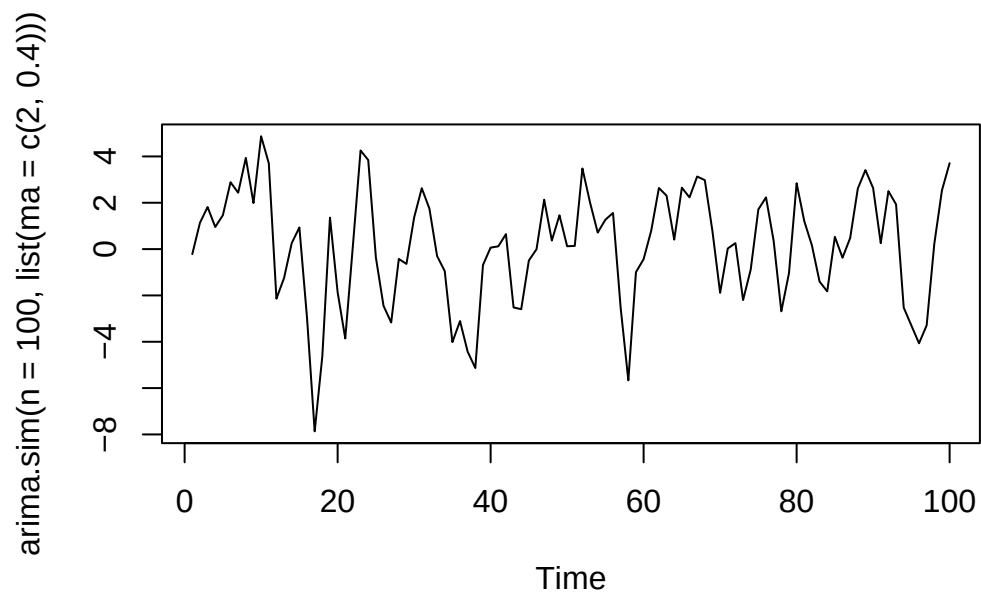
with the time series E_t of innovations.

i Note

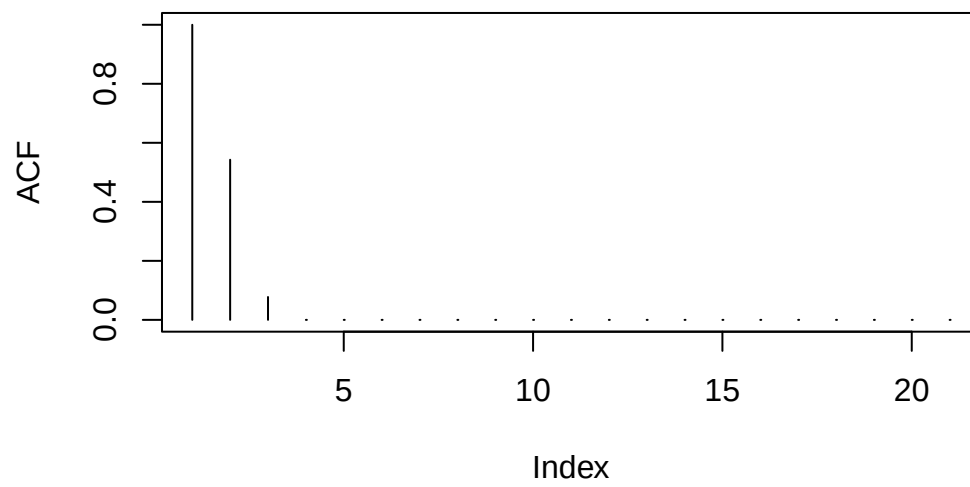
A moving average model does not have an explicit reference to previous values; only the innovation parts are considered.

```
set.seed(42)

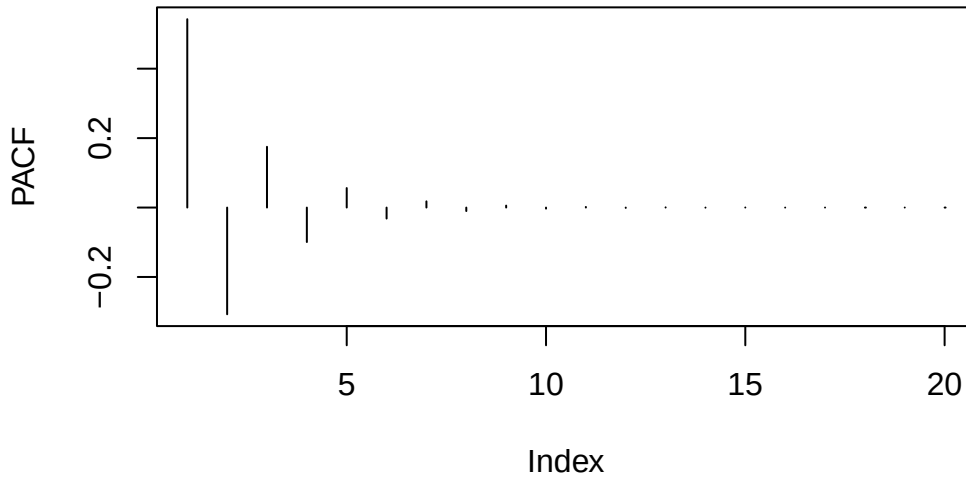
#MA(2)
plot(arima.sim(n=100,list(ma=c(2,0.4))))
```



```
plot(ARMAacf(ma=c(2,0.4),lag.max=20),type="h",ylab="ACF")
```



```
plot(ARMAacf(ma=c(2,0.4),pacf=T,lag.max=20),type="h",ylab="PACF")
```



Generalization: In general, we can show that the autocorrelation function of an $MA(q)$ vanishes for all lags larger than q .

W-06-Timeseries-Diagnostics-MA

Moving Average Model

A moving average $MA(q)$ model is a model in the form

$$X_t = E_t + \beta_1 E_{t-1} + \beta_2 E_{t-2} + \cdots + \beta_q E_{t-q}$$

with the time series E_t of innovations. The ACF displays cutoff and PACF displays fast decay. In general, we can show that the autocorrelation function of an $MA(q)$ vanishes for all lags larger than q .

Invertibility

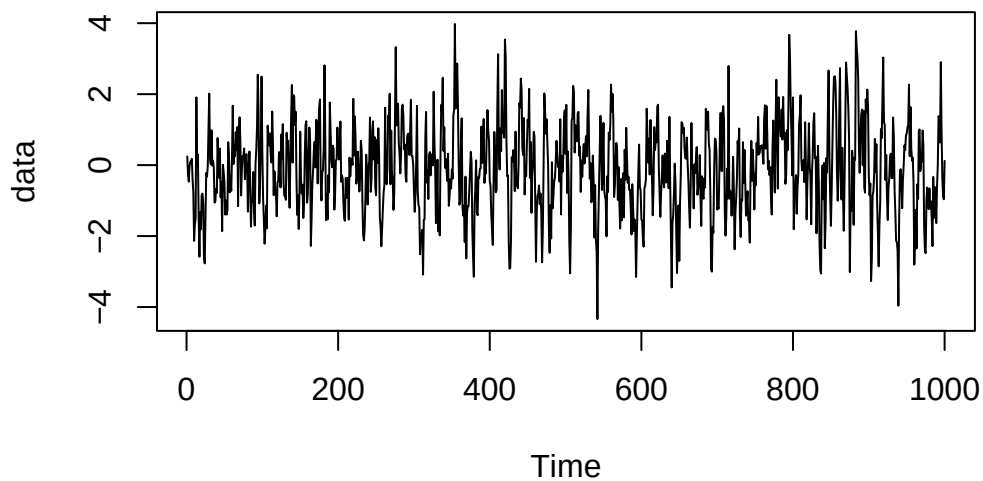
An $MA(q)$ model is invertible if the absolute value of the roots of the characteristic polynomial are larger than 1. Invertible $MA(q)$ processes can be expressed as $AR(\infty)$ processes.

Model Identification

$MA(q)$ models are (weakly) stationary, display clear cut-off in the ACF and PACF displays a fast decaying or a damped sinusoid behaviour.

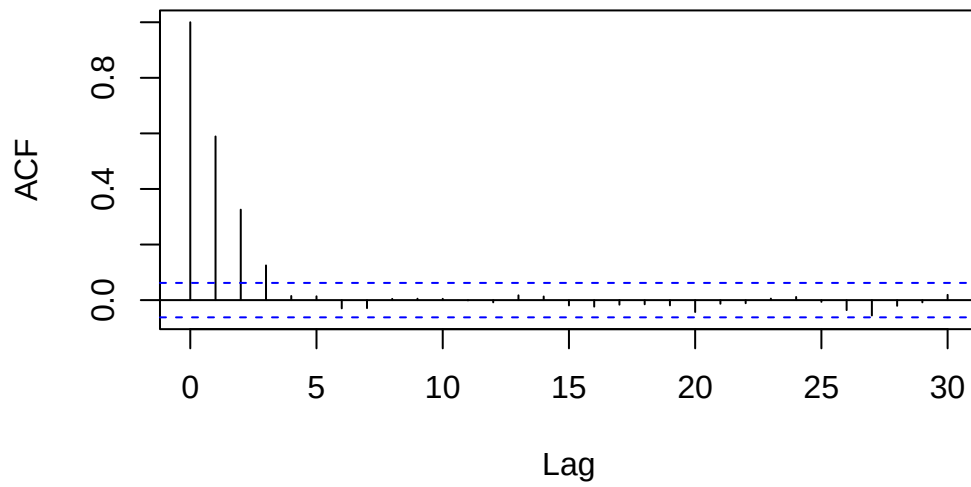
If these conditions are met, a $MA(q)$ model may be a good choice.

```
# Let us simulate an MA(3) process  
data<-arima.sim(n=1000,list(ma=c(0.6,0.4,0.2)))  
plot(data)
```



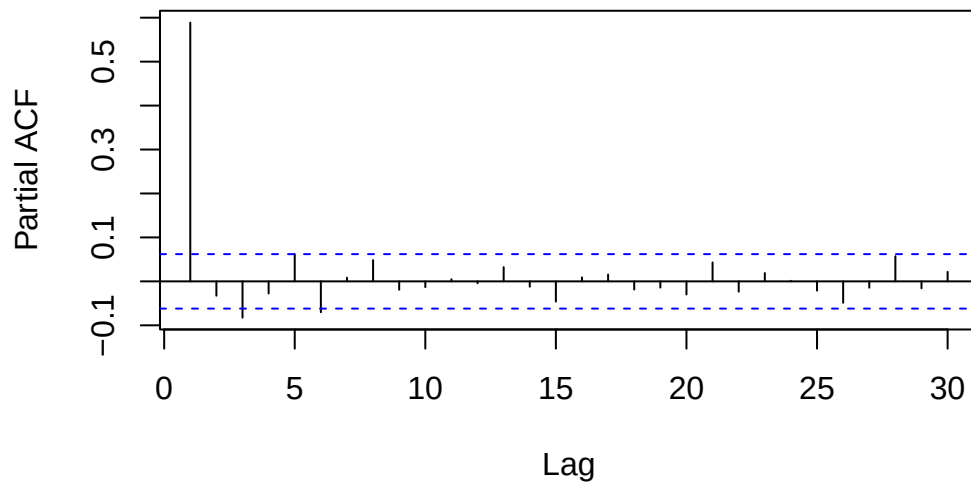
```
acf(data)
```


Series data



```
pacf(data)
```

Series data



Parameter Identification

The parameters of an $MA(q)$ model can be estimated with:

- Yule-Walker equations: As in the case of the $AR(p)$ processes, the parameters of $MA(q)$ processes can also be estimated by the Yule-Walker equations (after a global mean has been subtracted).
- Conditional sum of squares: `arima(data, order=c(...), method="CSS")`
- Maximum-likelihood estimation (default method in R): `arima(data, order=c(...), method="CSS-MLE")`

ARMA Models

An $ARMA(p, q)$ process combines an $AR(p)$ and a $MA(q)$ process

$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + E_t + \beta_1 E_{t-1} + \dots + \beta_q E_{t-q}$$

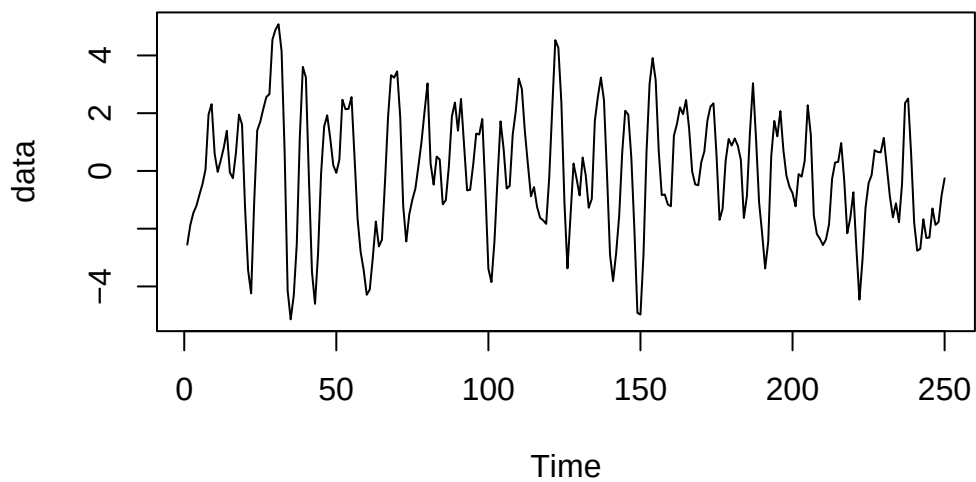
Definition

- Stationarity An $ARMA(p, q)$ model is stationary if the absolute value of the roots of characteristic polynomial is larger than 1. > Note: Analogy to the $AR(p)$ process
- Invertability An $ARMA(p, q)$ model is invertible if the absolute value of the roots of characteristic polynomial is larger than 1. > Note: Analogy to the $MA(q)$ process
- Consequence: Any stationary and invertible $ARMA(p, q)$ model can either be rewritten in the form of an $AR(\infty)$ or an $MA(\infty)$ model.

Note

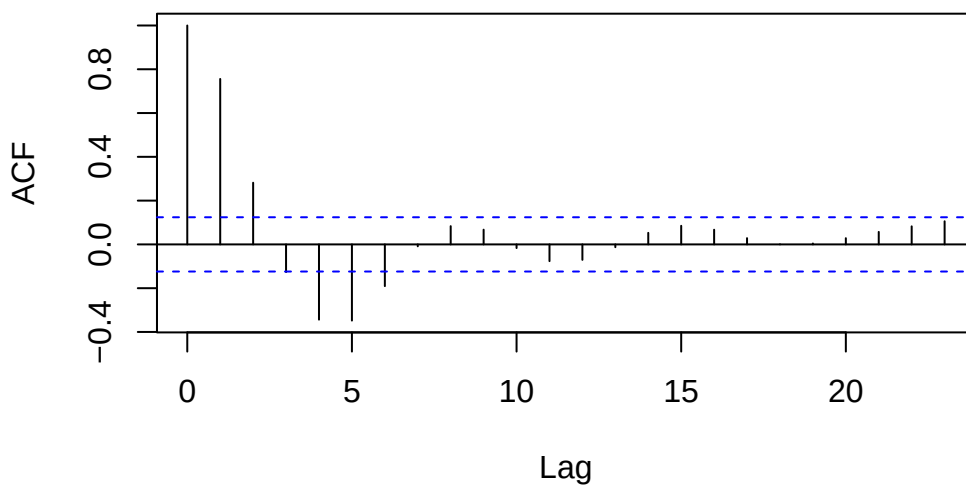
In practise $ARMA(p, q)$ models are preferred because they require fewer parameters.

```
data<-arima.sim(n=250,list(ar=c(1.1,-0.5),ma=c(0.4)))  
plot(data)
```

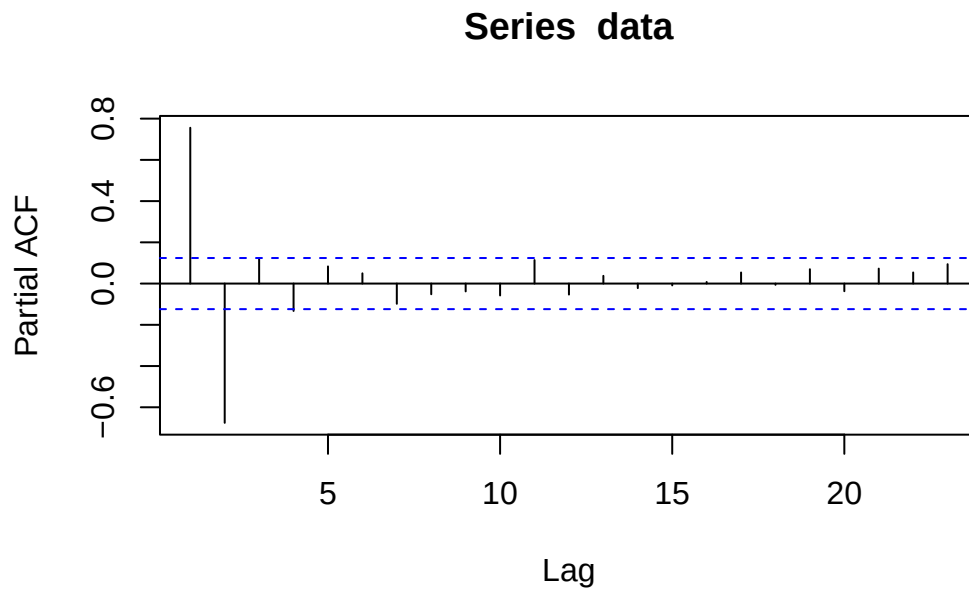


```
acf(data)
```

Series data



```
pacf(data)
```



Order Identification

Generic workflow for fitting:

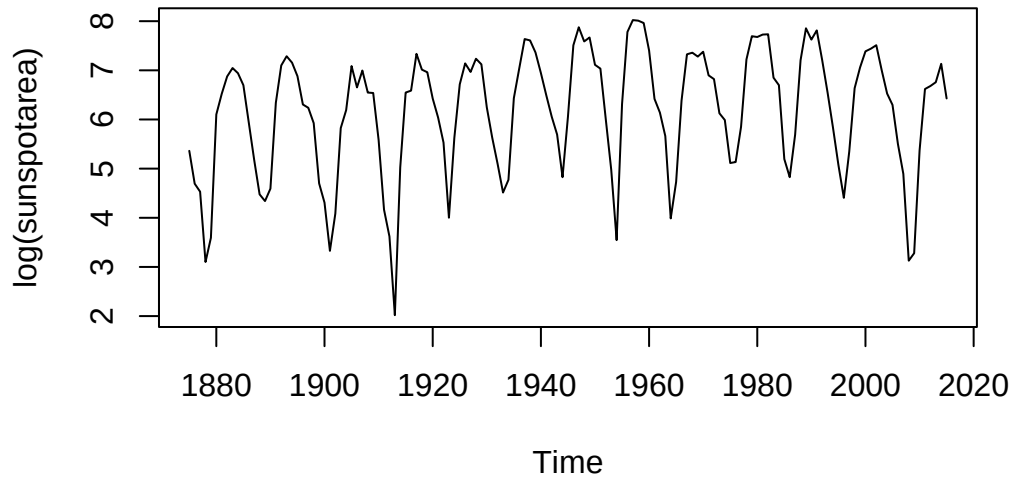
- Is the model stationary?
- Do the properties of (P)ACF match?
- Which orders p, q could be cut off in the (P)ACF?

Note: Even with theoretical (P)ACF the identification of the cut-off can be challenging.

Fitting ARMA Models

- Conditional sum of squares: Applicable for any invertible $ARMA(p, q)$ process
`arima(data, order=c(...), method=«CSS»)`
- Maximum likelihood estimation: Performs maximum likelihood estimation after CSS-process
`arima(..., order=c(...), method="CSS-ML")`

```
library(fpp2)
plot(log(sunspotarea))
```



```
library(forecast)
model <- auto.arima(log(sunspotarea), max.p = 10, max.q = 10, stationary = TRUE)
model
```

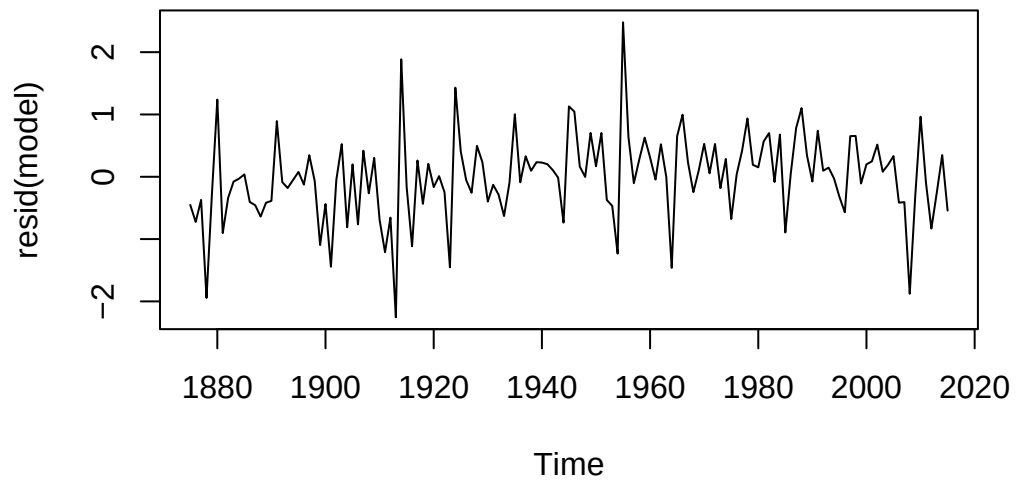
Series: log(sunspotarea)
ARIMA(2,0,1) with non-zero mean

Coefficients:

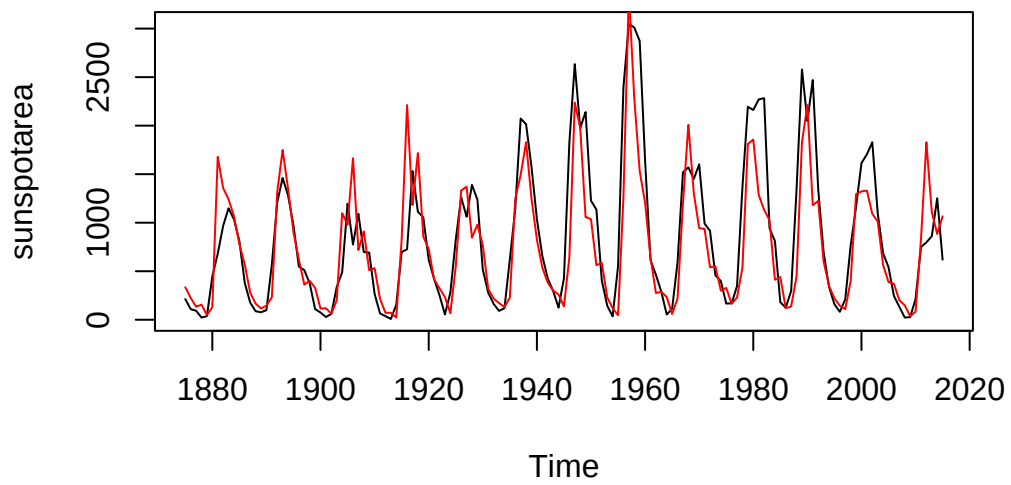
	ar1	ar2	ma1	mean
	1.4834	-0.7710	-0.5132	6.1885
s.e.	0.0728	0.0605	0.0953	0.0983

sigma^2 = 0.4794: log likelihood = -147.06
AIC=304.12 AICc=304.56 BIC=318.86

```
plot(resid(model))
```



```
plot(sunspotarea)  
lines(exp(log(sunspotarea)-resid(model)),col="red")
```



Advantages

Estimates all parameters above and provides error estimates.

Disadvantage

Works for any distribution of the innovation. For distributions that deviate strongly from Gaussian type, results may be unreliable.

Key Properties of Models

Property	AR(p)	MA(q)	ARMA(p,q)
ACF	Fast decay / damped sinusoid	Cut-off at q	Drop at q
PACF	Cut-off at p	Fast decay	Drop at p

Figure 10: Model Properties

W-07-Timeseries-ARMA-GLS

Linear Regression with timeseries

Principle

Model the target time series Y_t as a (linear) combination/function of the constituent, with an error term E_t and the realizations $x_{t1}, \dots, x_{tp}$ of the p time series. Major challenge: In real-world data, the errors can be strongly correlated.

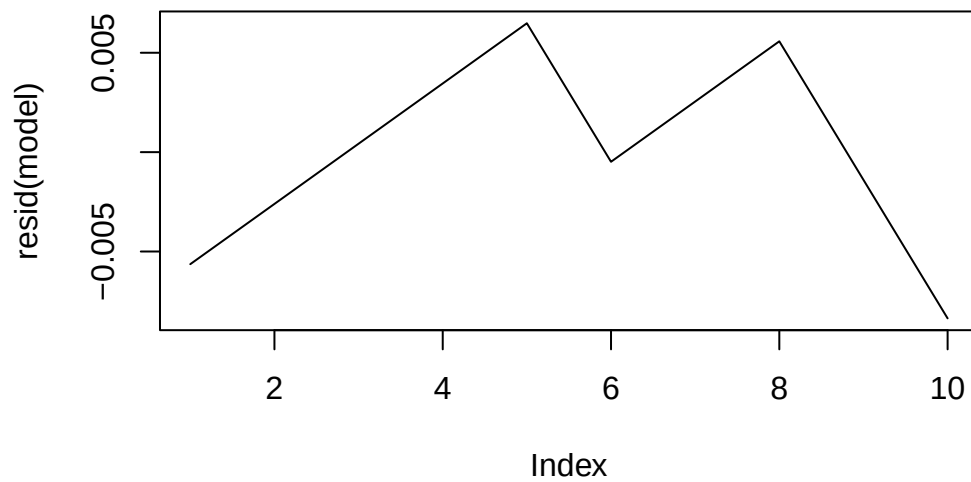
Find Model Coefficients

An assumption of linear regression is that the error E_t should be stationary and independent of the input time series. Often it is non-white noise and displays serial correlation. Perform a linear fit on the data and analyze the residuals:

```
# Sample Data
la_data <- data.frame(
  year = 2010:2019,
  value = c(3.79, 3.82, 3.85, 3.88, 3.91, 3.93, 3.96, 3.99, 4.01, 4.03)
)
```

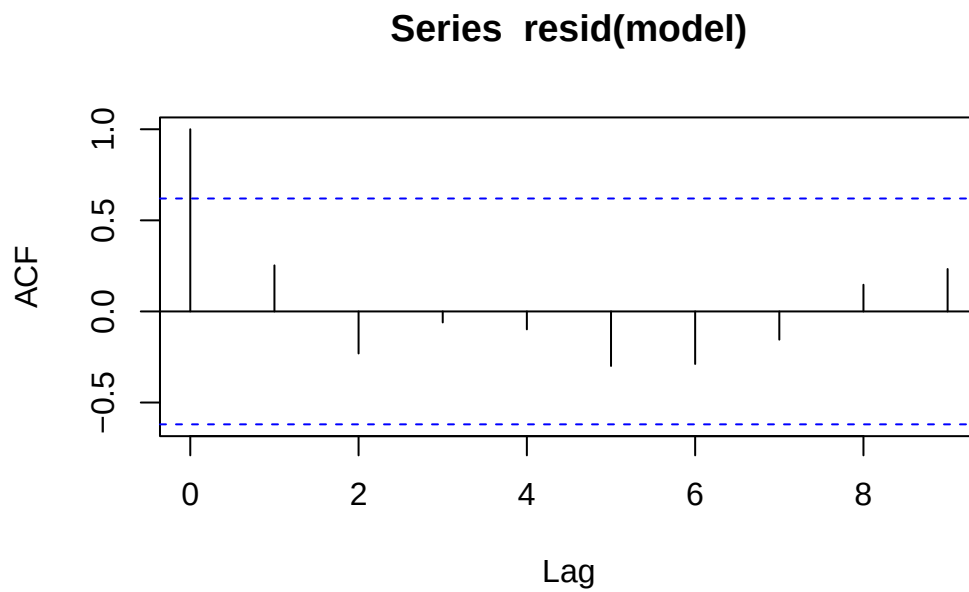
```
# Perform Linear Fit
model <- lm(value ~ year, data = la_data)

plot(resid(model), type="l")
```

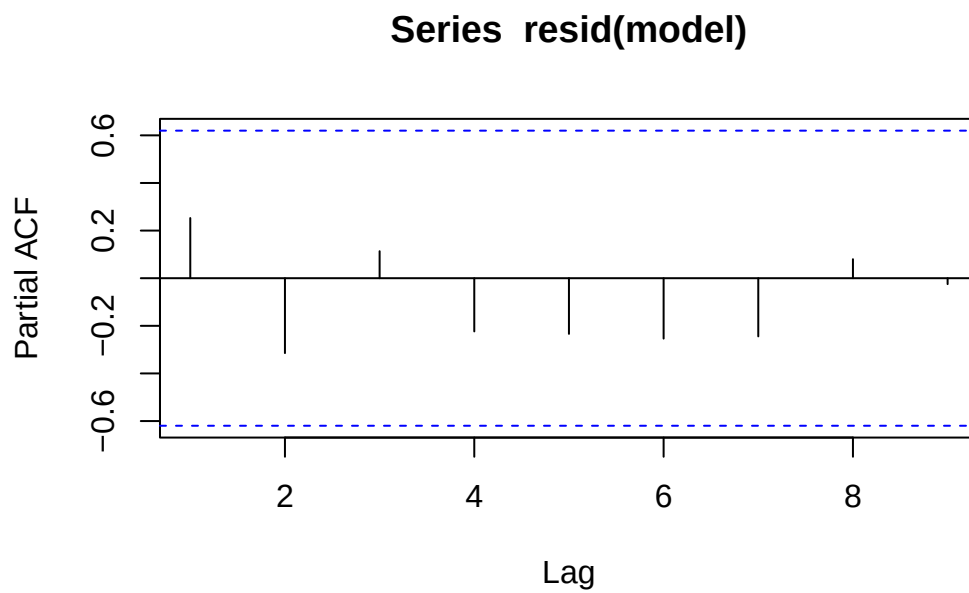


Is there a structure in the residuals?

```
acf(resid(model))
```

```
pacf(resid(model))
```



Error displays a serial connection and may be modeled by an AR(1) process. Whenever the error terms are dependent, the error estimation of the linear regression is invalid.

Note

Standard deviation is invalid if residuals are correlated

```
summary(model)
```

Call:

```
lm(formula = value ~ year, data = la_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0083636	-0.0023030	-0.0000303	0.0032273	0.0064848

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.041e+01	1.116e+00	-45.19	6.35e-11 ***
year	2.697e-02	5.538e-04	48.70	3.50e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.00503 on 8 degrees of freedom

Multiple R-squared: 0.9966, Adjusted R-squared: 0.9962

F-statistic: 2372 on 1 and 8 DF, p-value: 3.498e-11

Durbin-Watson Test

The Durbin-Watson (DW) test is a statistical test used to detect the presence of autocorrelation (specifically first-order serial correlation) in the residuals from a regression analysis.

In a standard linear regression, we assume that residuals are independent. If they are correlated (e.g., if the error at time t is related to the error at time $t - 1$), your standard errors may be underestimated, leading to misleadingly “significant” results.

```
library(lmtest)
dwtest(model)
```

Durbin-Watson test

data: model

DW = 0.99211, p-value = 0.009606

alternative hypothesis: true autocorrelation is greater than 0

Cochrane-Orcutt

The Cochrane-Orcutt method is an iterative procedure used in econometrics and statistics to estimate a linear regression model in the presence of autocorrelation (specifically AR(1) errors).

When the Durbin-Watson test indicates that residuals are correlated, the standard Ordinary Least Squares (OLS) assumptions are violated. Cochrane-Orcutt “cleanses” the model by transforming the variables to eliminate this correlation.

Steps

1. Perform ordinary least square fit on data.
2. Analyse (P)ACF and fit AR(1) model on residuals.
3. Transform data according to parameter of the residuals’ AR(1) process.
4. Fit the transformed data to determine the parameters of the linear model.

Generalized Least Square Methode

Generalized Least Squares (GLS) is a powerful extension of Ordinary Least Squares (OLS) used to estimate unknown parameters in a linear regression model when the underlying assumptions about the error terms are violated. Specifically, it is used when there is heteroscedasticity (non-constant variance) or autocorrelation (correlated errors).

Impact of Missing Variables

Correlated errors are often caused by missing input variables. In a perfect world, these parameters would be included in the model. However, often it is impossible to provide the input.

W-08-Timeseries-GLS-ARIMA

An ARIMA model is basically an ARMA model that has been “stretched” to include a trend. Because it contains that trend (the unit root), it doesn’t settle back to a constant average, making it non-stationary.

Definition

ARIMA stands for AutoRegressive Integrated Moving Average. It is a powerful and flexible class of statistical models used for analyzing and forecasting time-series data. Unlike simple linear regression, ARIMA explicitly accounts for the dependencies between observations over time. The model is defined by three parameters p, d, q .

Interpreting the Model Output

```
library(TSA)
data(oil.price)
fit<-arima(log(oil.price),order=c(2,1,1))
fit
```

Call:

```
arima(x = log(oil.price), order = c(2, 1, 1))
```

Coefficients:

	ar1	ar2	ma1
	-0.3240	0.0150	0.5981
s.e.	0.2653	0.1107	0.2571

sigma^2 estimated as 0.006642: log likelihood = 261.12, aic = -516.25

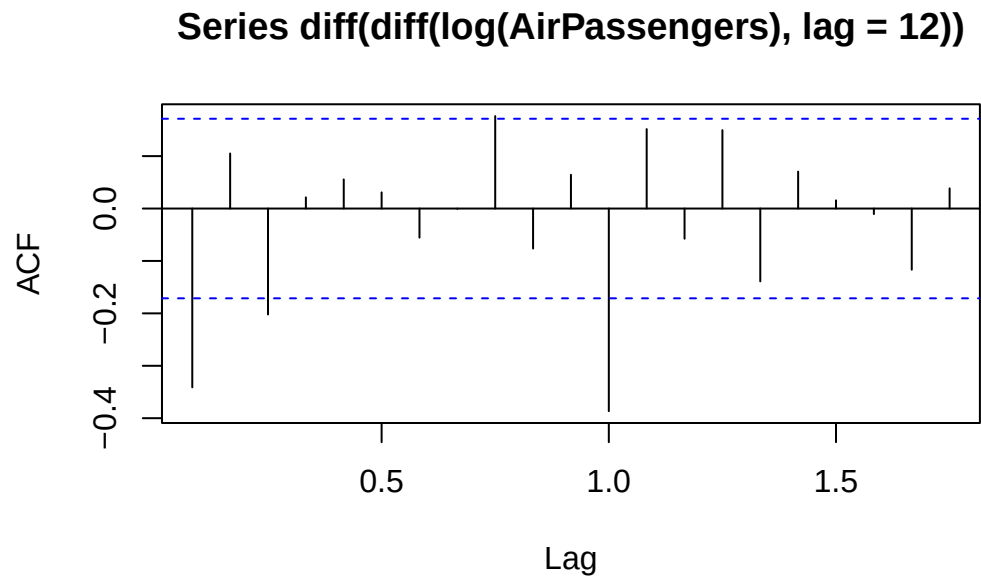
This ARIMA(2,1,1) model is equal to an ARMA(3,1) model.

W-09-Timeseries-ARIMA-SARIMA

Sarima Models

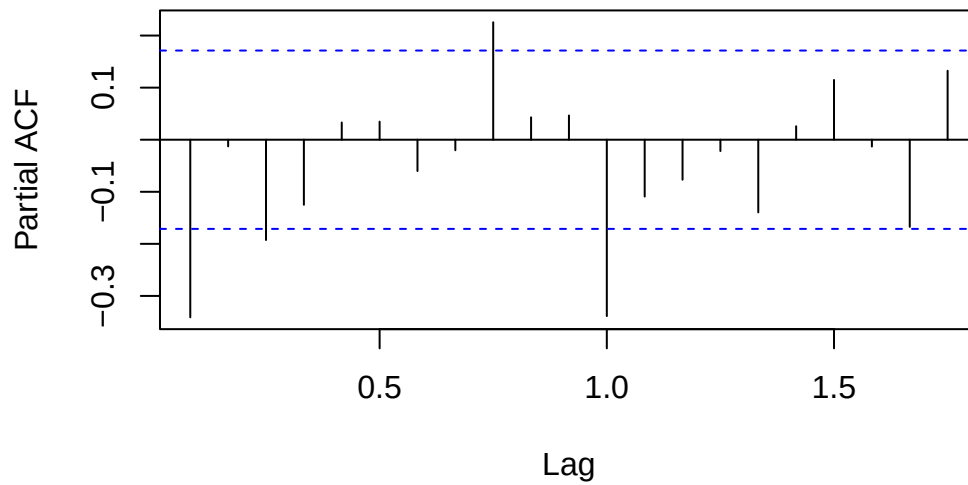
SARIMA stands for **S**easonal **A**utoregressive **I**ntegrated **M**oving **A**verage. It is an extension of the standard ARIMA model specifically designed to handle univariate time series data that contains a seasonal component. While ARIMA focuses on the relationship between points in time (trend), SARIMA adds a layer to account for cycles that repeat over specific intervals (e.g., monthly sales spikes or daily temperature changes).

```
data("AirPassengers")  
acf(diff(diff(log(AirPassengers),lag=12)))
```



```
pacf(diff(diff(log(AirPassengers),lag=12)))
```

Series diff(diff(log(AirPassengers), lag = 12))



```
fit<-arima(log(AirPassengers),order=c(0,1,1),seasonal=c(0,1,1))
fit
```

Call:

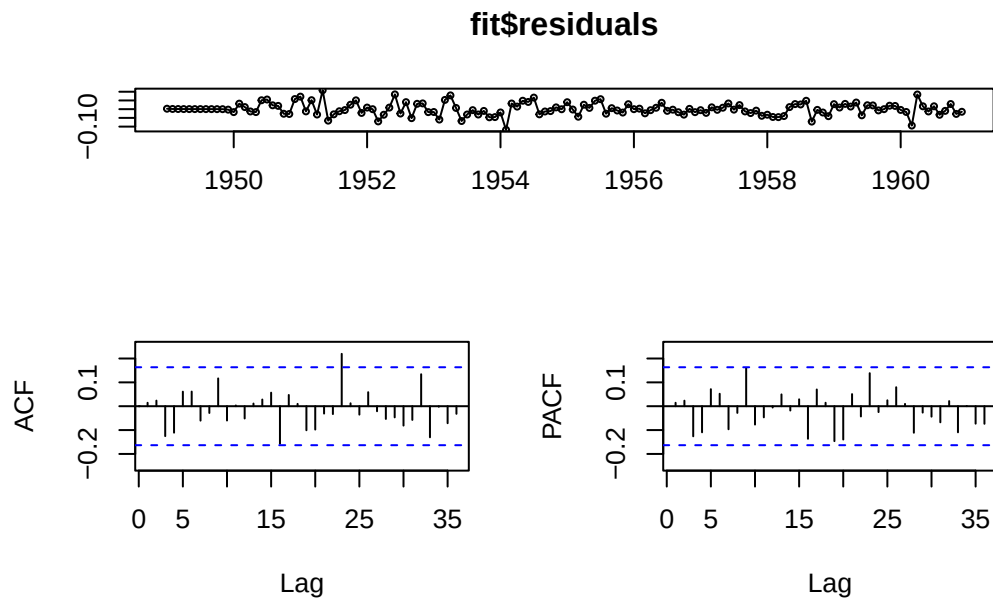
```
arima(x = log(AirPassengers), order = c(0, 1, 1), seasonal = c(0, 1, 1))
```

Coefficients:

	ma1	sma1
	-0.4018	-0.5569
s.e.	0.0896	0.0731

sigma^2 estimated as 0.001348: log likelihood = 244.7, aic = -485.4

```
library(forecast)
tsdisplay(fit$residuals)
```



```
fitAuto<-auto.arima(log(AirPassengers),ic="aic")
fitAuto
```

```
Series: log(AirPassengers)
ARIMA(0,1,1)(0,1,1)[12]
```

Coefficients:

	ma1	sma1
	-0.4018	-0.5569
s.e.	0.0896	0.0731

```
sigma^2 = 0.001371: log likelihood = 244.7
AIC=-483.4 AICc=-483.21 BIC=-474.77
```

Note: The automated procedure selects the same model.

How to fit SARIMA Models?

1. Estimate the seasonality period s visually. The order of differentiation D (for the seasonality part) is typically less or equal to 1.
2. Check if another differentiation of lag 1 is required for stationarity. If yes, try $d=1$, else $d=0$.

3. Analyse (P)ACF of the result of the two differentiations. Determine p,q from the short-lag behavior and P,Q from the multiple-of-the-period dependency.
4. Fit the model with `arima(data,order=c(p,d,q),seasonal=c(P,D,Q))`
5. Check the applicability of the model by analyzing the residuals (residuals should behave like (Gaussian) white noise).

Non-Linear Models

Linear models are models that can be formulated as linear combination of different time series elements X_t and errors/innovations E_t .

Note: An example are data with changing standard deviation.

Heteroscedasticity

Heteroscedasticity occurs when the variance of the residuals (errors) in a model is not constant across all levels of the independent variables or across time.

Note: Heteroskedastic time series are not stationary, because the variance is not equal for all time points.

ARCH Models

1. Overview

ARCH stands for **Autoregressive Conditional Heteroskedasticity**. Developed by Robert Engle in 1982, these models are specifically designed to model and forecast time-varying volatility (risk) in financial and economic time series data.

Unlike traditional time series models (like ARIMA) which assume constant variance (homoskedasticity), ARCH models capture the phenomenon of **volatility clustering**—the tendency for large shocks to be followed by large shocks, and small shocks to be followed by small shocks.

2. Core Concepts

- **Heteroskedasticity:** The variance of the error terms is not constant over time.
- **Conditional:** The volatility at the current time step depends on past information.
- **Autoregressive:** The model predicts current variance by looking back at its own past squared error terms.

3. Mathematical Formulation

An $ARCH(p)$ model consists of two parts: the **mean equation** and the **variance equation**.

The Mean Equation

The observed return or value r_t is modeled as a function of an expected value μ_t (which can be a constant or an ARMA process) and a residual error term ϵ_t :

$$r_t = \mu_t + \epsilon_t$$

Where the error term ϵ_t is broken down into a standard normal random variable z_t and the conditional standard deviation σ_t :

$$\epsilon_t = \sigma_t z_t \quad \text{where} \quad z_t \sim \mathcal{N}(0, 1)$$

The Variance Equation (ARCH(p))

The conditional variance σ_t^2 is modeled as a linear function of p lagged squared residuals from the past:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2 + \dots + \alpha_p \epsilon_{t-p}^2$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2$$

Constraints for stability and positivity: * $\alpha_0 > 0$ (Ensures variance is always strictly positive) * $\alpha_i \geq 0$ for all $i > 0$ (Ensures non-negative contributions) * $\sum_{i=1}^p \alpha_i < 1$ (Ensures the process is stationary)

4. Why Use ARCH Models?

- **Risk Management:** Essential for calculating dynamic Value-at-Risk (VaR) and Expected Shortfall.
- **Asset Pricing:** Crucial for pricing options and other derivatives where volatility is a key input.
- **Realistic Forecasting:** It expands standard confidence intervals during turbulent market periods, preventing underestimation of risk.

5. Extension: The GARCH Model

In practice, pure $ARCH(p)$ models often require a large number of lags (p) to capture financial volatility accurately. To solve this, Tim Bollerslev introduced the **GARCH** (Generalized ARCH) model, which includes lagged values of the variance itself:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

A standard **GARCH(1,1)** model is typically sufficient to model most financial asset returns effectively.

A time series is autoregressive conditional heteroskedastic of the order p (abbreviated ARCH(p)) if it has the form:

$$X_t = \sigma_t W_t \quad \text{with} \quad \sigma_t = \sqrt{\alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2}$$

Fitting ARCH Models

```
#load SMI data
data("EuStockMarkets")

esm <- EuStockMarkets
tmp <- EuStockMarkets[,2]
smi <- ts(tmp, start=start(esm), freq=frequency(esm))
lret.smi <- diff(log(smi))

#perform fit
library(tseries)
fit<-garch(lret.smi, order=c(0,2))
```

***** ESTIMATION WITH ANALYTICAL GRADIENT *****

I	INITIAL X(I)	D(I)
1	7.700685e-05	1.000e+00
2	5.000000e-02	1.000e+00

```
3      5.000000e-02      1.000e+00
```

IT	NF	F	RELDF	PRELDF	RELDX	STPPAR	D*STEP	NPRELDF
0	1	-7.796e+03						
1	8	-7.797e+03	7.76e-05	1.39e-04	2.7e-05	1.4e+11	2.7e-06	9.94e+06
2	9	-7.797e+03	7.84e-07	8.41e-07	2.1e-05	2.0e+00	2.7e-06	3.10e+00
3	17	-7.803e+03	8.48e-04	1.27e-03	2.6e-01	2.0e+00	4.5e-02	3.10e+00
4	19	-7.803e+03	5.19e-06	7.74e-06	2.0e-02	2.0e+00	4.5e-03	1.44e-03
5	20	-7.803e+03	2.15e-07	2.70e-06	2.0e-02	2.0e+00	4.5e-03	3.72e-04
6	21	-7.803e+03	3.22e-06	2.03e-05	1.3e-02	1.9e+00	2.2e-03	3.91e-04
7	24	-7.805e+03	1.73e-04	1.62e-04	9.7e-02	4.9e-01	2.1e-02	1.87e-04
8	25	-7.805e+03	9.03e-05	1.05e-04	1.6e-01	0.0e+00	3.8e-02	1.05e-04
9	26	-7.805e+03	1.19e-05	1.35e-05	4.7e-02	0.0e+00	1.3e-02	1.35e-05
10	27	-7.805e+03	1.16e-07	1.07e-07	4.1e-03	0.0e+00	1.1e-03	1.07e-07
11	28	-7.805e+03	1.13e-09	1.10e-09	3.2e-04	0.0e+00	1.0e-04	1.10e-09
12	29	-7.805e+03	8.85e-12	7.83e-12	4.0e-05	0.0e+00	1.1e-05	7.83e-12

***** RELATIVE FUNCTION CONVERGENCE *****

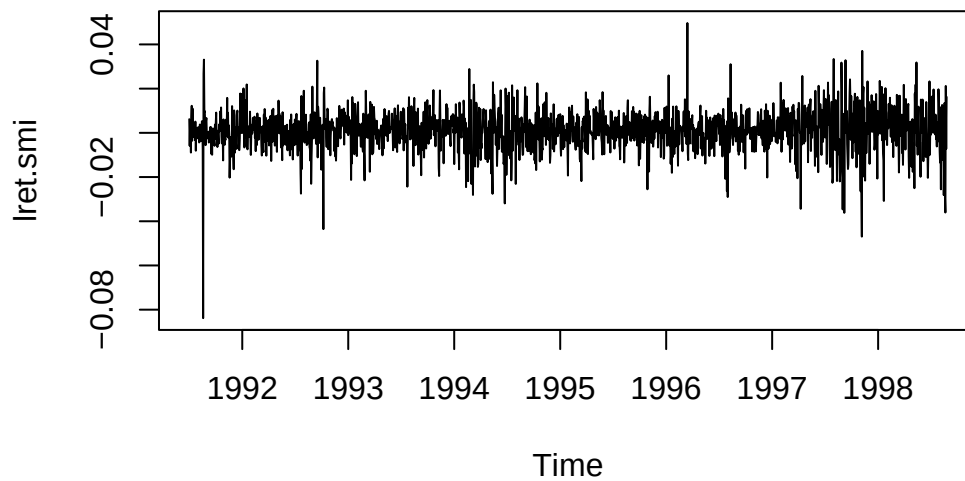
FUNCTION	-7.805450e+03	RELDX	3.982e-05
FUNC. EVALS	29	GRAD. EVALS	13
PRELDF	7.826e-12	NPRELDF	7.826e-12

I	FINAL X(I)	D(I)	G(I)
1	6.567582e-05	1.000e+00	1.512e+00
2	1.309070e-01	1.000e+00	1.394e-03
3	1.074336e-01	1.000e+00	-8.848e-04

```
fit$coef
```

a0	a1	a2
6.567582e-05	1.309070e-01	1.074336e-01

```
plot(lret.smi)
```



```
fit$fitted.values
```

Time Series:

Start = c(1991, 131)

End = c(1998, 169)

Frequency = 260

	sigt	-sigt
1991.500	NA	NA
1991.504	NA	NA
1991.508	0.008619948	-0.008619948
1991.512	0.008413776	-0.008413776
1991.515	0.008192281	-0.008192281
1991.519	0.008738385	-0.008738385
1991.523	0.008954976	-0.008954976
1991.527	0.009492994	-0.009492994
1991.531	0.009133273	-0.009133273
1991.535	0.009107082	-0.009107082
1991.538	0.009010984	-0.009010984
1991.542	0.008347955	-0.008347955
1991.546	0.008228813	-0.008228813

1991.550 0.008218464 -0.008218464
1991.554 0.008182066 -0.008182066
1991.558 0.008192224 -0.008192224
1991.562 0.008379568 -0.008379568
1991.565 0.008776044 -0.008776044
1991.569 0.008527705 -0.008527705
1991.573 0.008162093 -0.008162093
1991.577 0.008164387 -0.008164387
1991.581 0.008160428 -0.008160428
1991.585 0.008151230 -0.008151230
1991.588 0.008112088 -0.008112088
1991.592 0.008156176 -0.008156176
1991.596 0.008296850 -0.008296850
1991.600 0.008366156 -0.008366156
1991.604 0.008332257 -0.008332257
1991.608 0.008198592 -0.008198592
1991.612 0.008183117 -0.008183117
1991.615 0.008274863 -0.008274863
1991.619 0.008207454 -0.008207454
1991.623 0.008337280 -0.008337280
1991.627 0.008361690 -0.008361690
1991.631 0.008594171 -0.008594171
1991.635 0.031486098 -0.031486098
1991.638 0.030248146 -0.030248146
1991.642 0.016931529 -0.016931529
1991.646 0.013959892 -0.013959892
1991.650 0.009355178 -0.009355178
1991.654 0.008882279 -0.008882279
1991.658 0.008496063 -0.008496063
1991.662 0.008529784 -0.008529784
1991.665 0.008638284 -0.008638284
1991.669 0.008397910 -0.008397910
1991.673 0.008152679 -0.008152679
1991.677 0.008146693 -0.008146693
1991.681 0.008157023 -0.008157023
1991.685 0.008147135 -0.008147135
1991.688 0.008261294 -0.008261294
1991.692 0.008293718 -0.008293718
1991.696 0.008277651 -0.008277651
1991.700 0.008255187 -0.008255187
1991.704 0.008146510 -0.008146510
1991.708 0.008159992 -0.008159992
1991.712 0.008849384 -0.008849384

1991.715 0.008685582 -0.008685582
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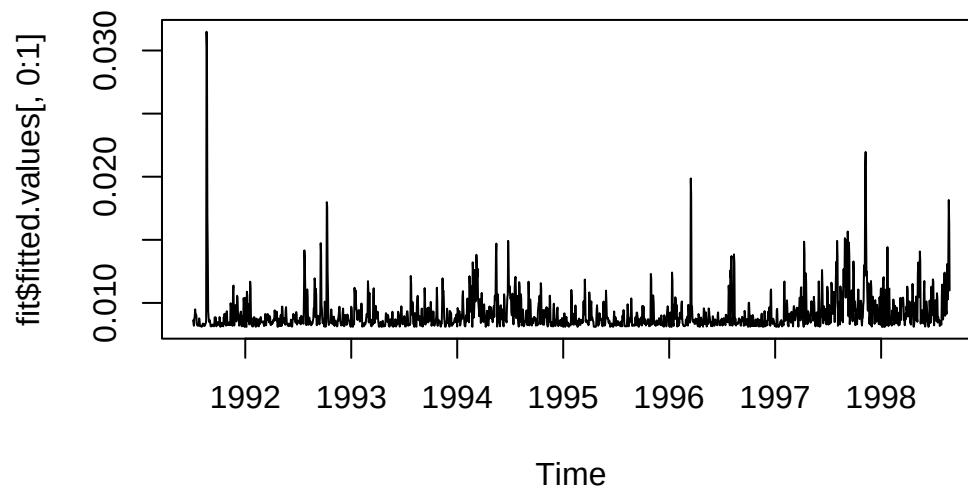
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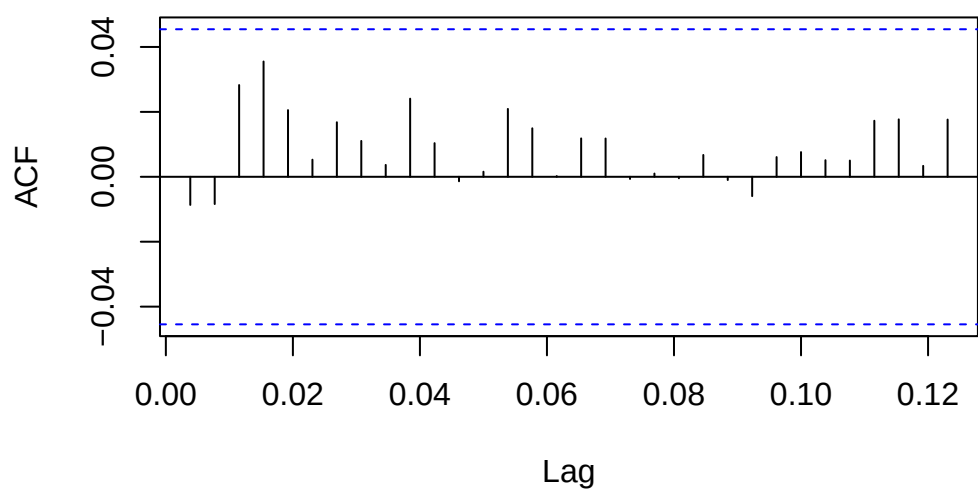
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plot(fit$fitted.values[,0:1])
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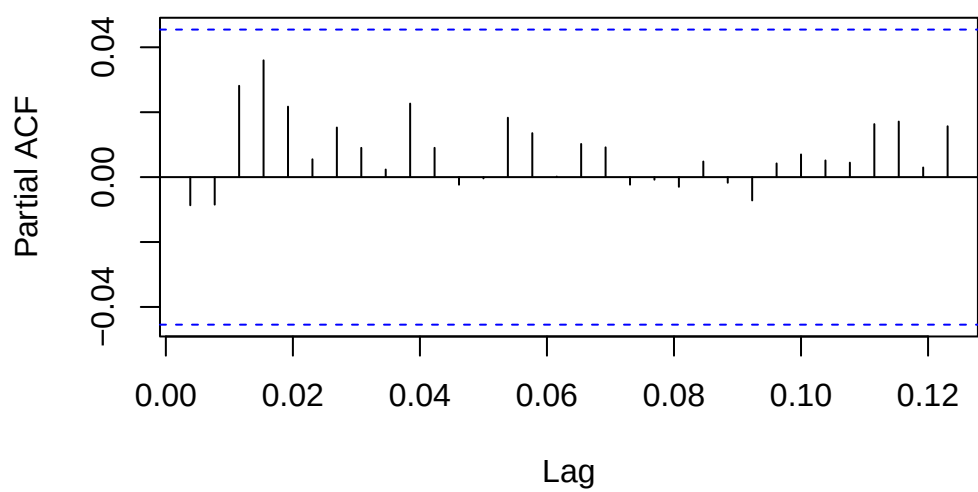
```
values<-fit$residuals  
acf(values^2,na.action=na.pass)
```


Series values²



```
pacf(values2,na.action=na.pass)
```

Series values²



Fundamentals of Forecasting

Generally speaking, we are looking for the most likely next k values based on the first n observations.

Sources of Uncertainty in Forecasting

Principal sources of uncertainty:

1. Is the model of the past data also appropriate for the future (e.g. change of behavior, policy)?
2. Did you choose the order of the ARMA(p, q) correctly?
3. Are the fitting parameters in the model (sufficiently) accurate?
4. How can you assess the contribution/variability of the innovation terms E_t ?

Forecasting different Models

$AR(1)$ processes

For an $AR(1)$ model the forecasted value is just the last value times the fitting coefficient α_1 .

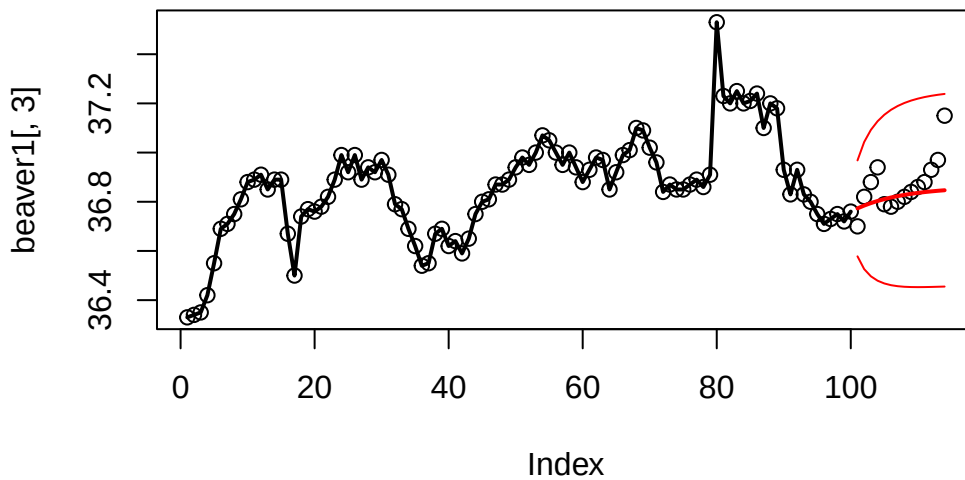
Note: Only the previous value is required.

```
#Extract training set
btrain<-window(beaver1[,3],1,100)

#Fit AR(1) model with Burg's algorithm
fit<-ar.burg(btrain,order=1)

#Perform forecast of 14 steps
forecast <- predict(fit, n.ahead=14)

#Plot forecast
plot(beaver1[,3], lty=3)
lines(btrain, lwd=2)
lines(forecast$pred, lwd=2, col="red")
lines(forecast$pred+forecast$se*1.96, col="red")
lines(forecast$pred-forecast$se*1.96, col="red")
```



$AR(p)$ models

For an $AR(p)$ model the p -step forecast is k steps ahead, the formula is calculated with the previous predictions. The prognosis intervals only consider the uncertainty of the (Gaussian) innovation. Other sources are neglected.

Forecastin *ARIMA* Models

Mathematically derivation of prediction more involved. However, everything is implemented in the predict function.

```
#Read data
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/foehre.dat"
douglasfir <- ts(scan(t.url, skip=1), start=1107)
train <- window(douglasfir, start=1107, end=1900)

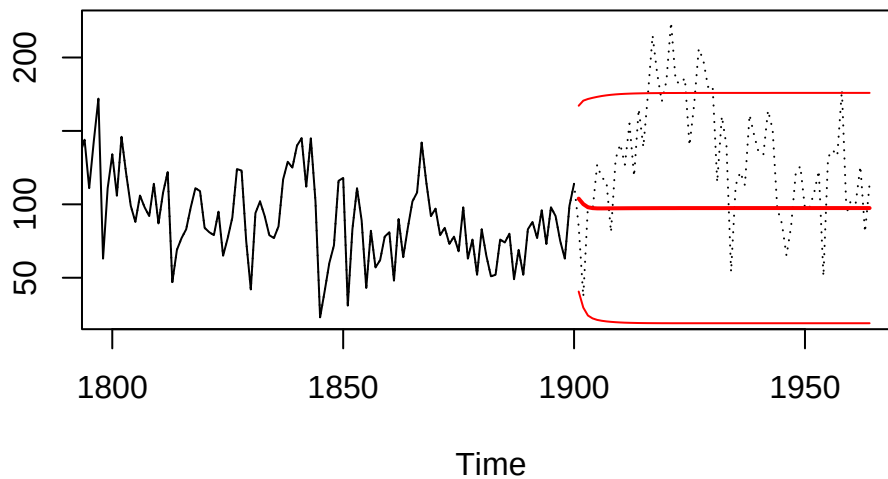
#Generate ARMA(4,1) model
fit <- arima(train, order=c(4,0,1))

# Predict 64 steps
fc <- predict(fit, n.ahead=64)
# Plot result
```

```

plot(window(douglasfir, 1800, 1964), lty=3, ylab="")
lines(train, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")

```



Forecast for time series with trend and seasonality

If data are fit by an *SARIMA* model, R takes care of all aspects of the forecast.

```

#Load data
lap<-log(AirPassengers)
ltrain<-window(lap,1949,1959)

#Perform fit
fit<-arima(ltrain,order=c(0,1,1),seasonal=c(0,1,1))

#Perform forecast
fc <- predict(fit, n.ahead=24)

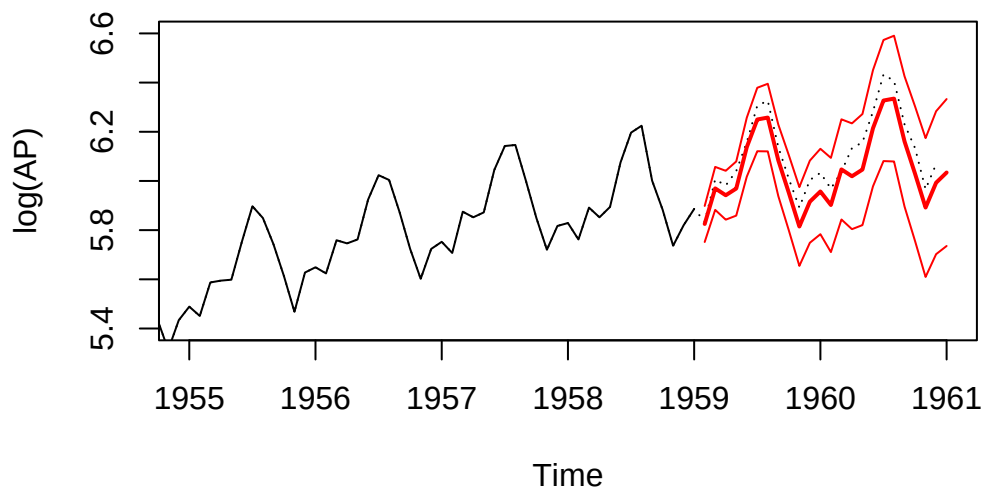
#Plot everything

```

```

plot(lap, lty=3,
     xlim=c(1955,1961),ylim=c(5.4,6.6),ylab="log(AP)")
lines(ltrain, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")

```



Features

- The predicted values capture the trend and seasonality.
- The prediction intervals will typically increase over time.
- The prediction quality of the trend is not ascertained and may be poor.

W-10-Timeseries-Forecasting

Forecasting MA, AR and ARMA Models

For an $AR(1)$ model $x_t = \alpha_1 Y_{t-1} + E_t$ the 1-step forecast can be calculated as:

$$\hat{X}_{n+1;1:n} = \alpha_1 x_n$$

i.e. the forecasted value is just the last value times the fitting coefficient.

Note: Here, only the last value is required.

The 95% prognosis interval is then (for Gaussian innovations) is $\alpha_1 x_n \pm 1.96\sigma_e$. The prognosis intervals only consider the uncertainty of the (Gaussian) innovation. Other sources are neglected.

For an AR(p) model, the p-step forecast is:

$$\hat{X}_{n+1;1:n} = \alpha_1 x_n + \dots + \alpha_p x_{n+1-p}$$

i Note

For k steps ahead, the formula is calculated with the previous predictions.

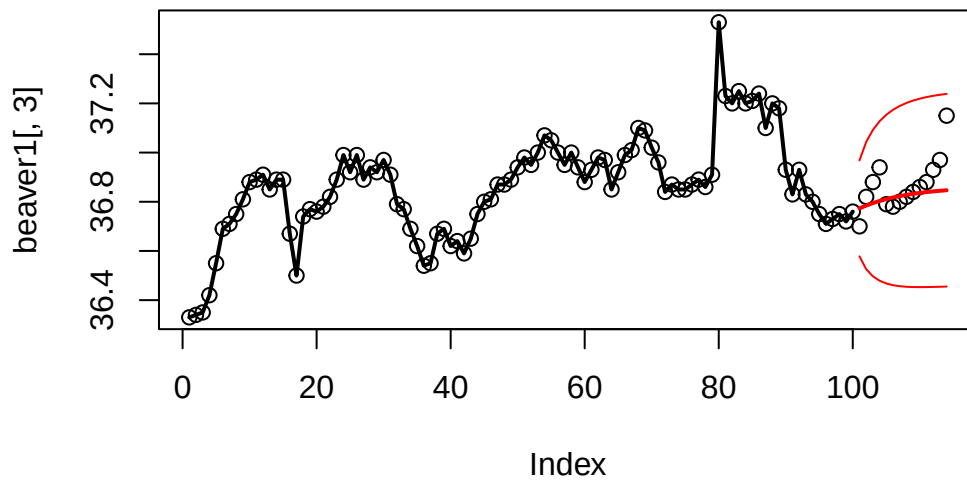
```
data(beavers)

#Extract training set
btrain<-window(beaver1[,3],1,100)

#Fit AR(1) model with Burg's algorithm
fit<-ar.burg(btrain,order=1)

#Perform forecast of 14 steps
forecast <- predict(fit, n.ahead=14)

#Plot forecast
plot(beaver1[,3], lty=3)
lines(btrain, lwd=2)
lines(forecast$pred, lwd=2, col="red")
lines(forecast$pred+forecast$se*1.96, col="red")
lines(forecast$pred-forecast$se*1.96, col="red")
```



Forecasting ARIMA Models

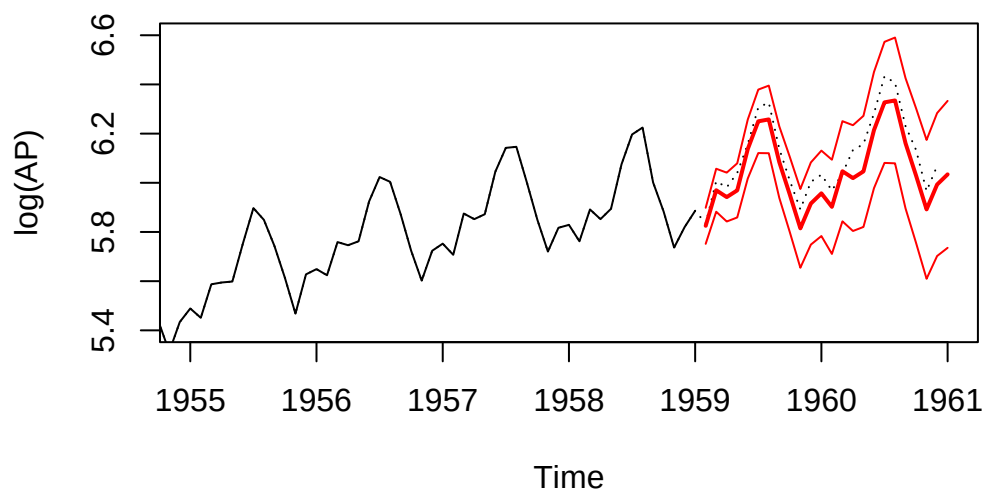
The predicted values capture the trend and seasonality. The prediction intervals will typically increase over time. The prediction quality of the trend is not ascertained and may be poor.

```
#Load data
lap<-log(AirPassengers)
ltrain<-window(lap,1949,1959)

#Perform fit
fit<-arima(ltrain,order=c(0,1,1),seasonal=c(0,1,1))

#Perform forecast
fc <- predict(fit, n.ahead=24)

#Plot everything
plot(lap, lty=3,
xlim=c(1955,1961),ylim=c(5.4,6.6),ylab="log(AP)")
lines(ltrain, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")
```



Forecasting of Decomposed Models

Before introducing SARIMA models, we established the following workflow:

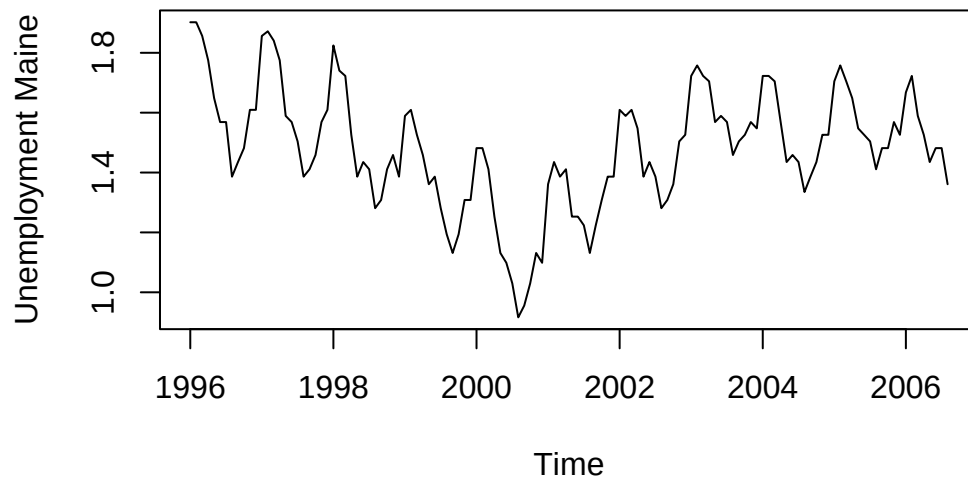
1. Remove trend
2. Remove seasonality effects
3. Fit the remainder with an ARMA(p,q) model

How to use this decomposition for a forecast?

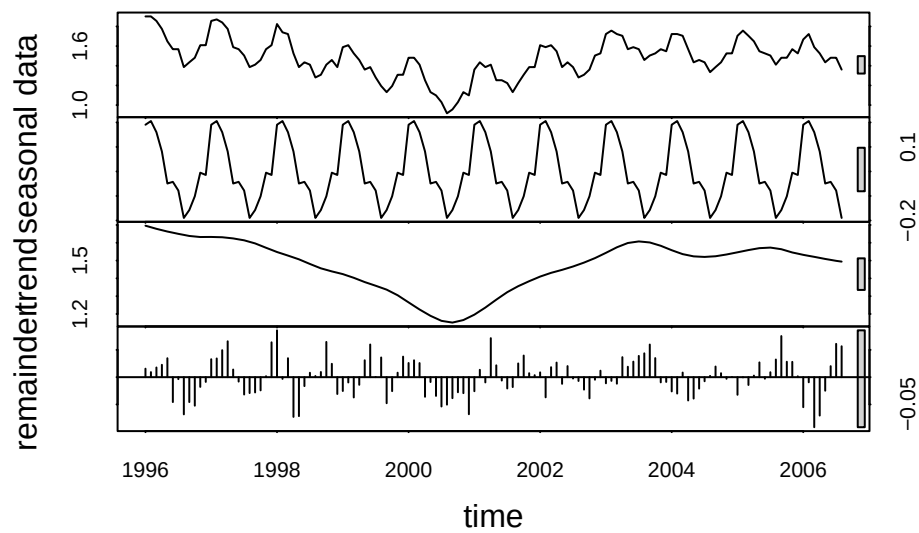
1. Trend: Extrapolate linearly or manually
2. Seasonal pattern: Copy previous values or extrapolate
3. Stationary remainder: Use forecast functions.

```
address <- "https://raw.githubusercontent.com/prabeshdhakal/Introductory-Time-Series-with-R-1
dat <- read.table(address,header=T)
tsd <- ts(dat,start=c(1996,1),freq=12)

plot(log(tsd[,1]),ylab="Unemployment Maine")
```

```
#Decompose data in trend, seasonal and remainder
decomp<-stl(log(tsd[,1]),s.window="periodic")
plot(decomp)
```



1. Trend Forecast

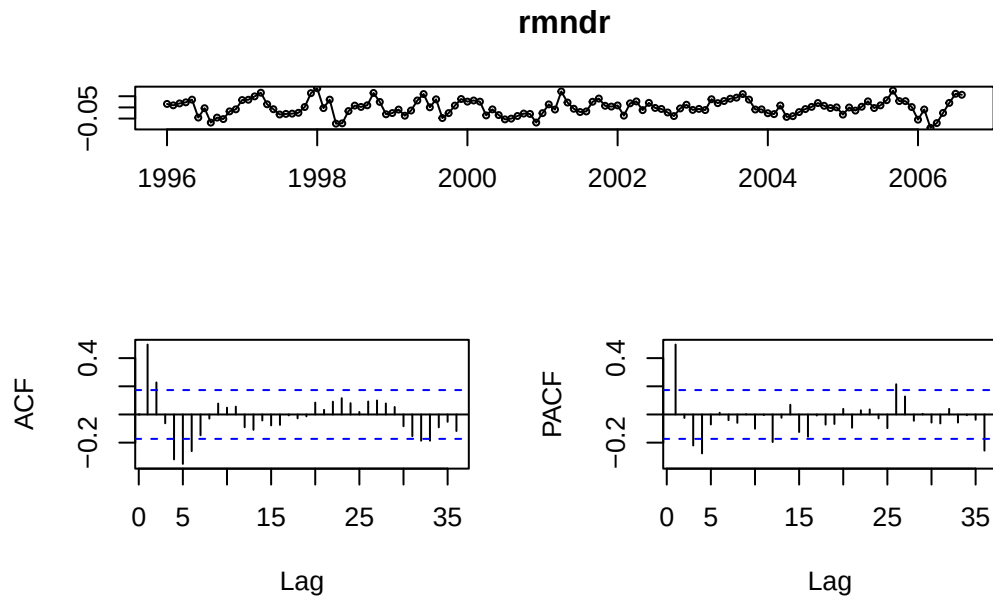
```
# Trend forecast by linear extrapolation
# Actual interpolation
xx <- time(decomp$time.series[,2])[105:128]
yy <- decomp$time.series[105:128,2]
fit.regr <- lm(yy~xx)
t.fore <- 1.494 + (0:23)/12 * coef(fit.regr)[2]
```

2. Seasonal Forecast

```
# Seasonal forecast using last values
season <- decomp$time.series[,1]
l2y <- window(season,start=c(2004,9),end=c(2006,8))
s.fore <- ts(l2y, start=c(2006,9), end=c(2008,8), freq=12)
```

3. Fitting the remainder

```
rmndr <- decomp$time.series[,3]
tsdisplay(rmndr)
```



```
fit.rmndr <- arima(rmndr, order=c(4,0,0), include.mean=F)
r.fore <- predict(fit.rmndr, n.ahead=24)$pred
```

4. Final Forecast

```
# Adding the 3 components
fore <- t.fore + s.fore + r.fore
```

Panel Data

The defining characteristic of panel data is that it combines two dimensions: cross-sectional data and time-series data. In short: you observe the same subjects (entities) repeatedly over multiple time periods.

Sequence Analytics

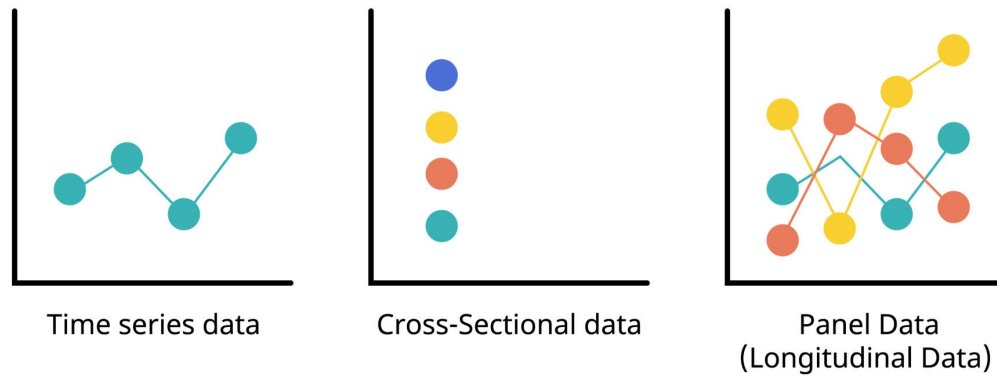


Figure 11: Panel Data

Balanced Panel Data

every single subject/entity is observed for the exact same time periods. There are no missing slots in your matrix.

If you have N subjects and T time periods, a perfectly balanced panel will have exactly $N \times T$ observations.

Note: Unbalanced panel = A panel data set that is not balanced.

Fixed Effect Models

A **Fixed Effects (FE) Model** is primarily used with **panel data** (longitudinal data), which observes multiple entities (e.g., individuals, companies, countries) across multiple time periods.

Core Concept

It controls for **unobserved heterogeneity** by factoring out all individual characteristics that do not change over time (time-invariant variables). This eliminates omitted variable bias caused by these hidden factors. The model solely analyzes the “within-variation” (how variables change within the *same* entity over time).

The Basic Equation

$$Y_{it} = \beta * X_{it} + \alpha_i + u_{it}$$

- **Y_it**: Dependent variable for entity i at time t .
- **X_it**: Time-varying independent variable(s).
- **beta**: The effect of the independent variable(s) we want to estimate.
- **alpha_i**: The **Fixed Effect** (an individual intercept that captures all time-invariant traits of entity i).
- **u_it**: Error term (time-varying unobserved factors).

Time Fixed Effects

Besides individual effects, you can include **Time Fixed Effects** to control for macro-events that affect all entities equally at a specific point in time (e.g., a global financial crisis). Using both individual and time effects is called a **Two-Way Fixed Effects Model**.

FE vs. Random Effects (RE)

- **Fixed Effects**: Assumes unobserved individual traits *correlate* with the independent variables.
- **Random Effects**: Assumes unobserved individual traits do *not* correlate with the independent variables.
- *Note*: The **Hausman Test** is typically used to statistically determine which model is more appropriate.

Pros & Cons

Advantages: * **Reduces Bias:** Elegantly solves the omitted variable bias for any unobserved, time-invariant factors. * **Causality:** Provides a stronger case for causal inference compared to simple cross-sectional models.

Disadvantages: * **No Time-Invariant Variables:** You cannot estimate the effects of variables that don't change over time (like gender or birthplace), as they are absorbed by the fixed effect (α_i). * **Low Efficiency with Low Variation:** If your independent variables barely change over time, the model loses statistical power and becomes imprecise. * **Degrees of Freedom:** In panels with many entities, the model consumes a massive amount of degrees of freedom to estimate all the individual intercepts.

What if each entity has its bias?

A global least squares fit may yield wrong models for data with a constant offset between two entities.

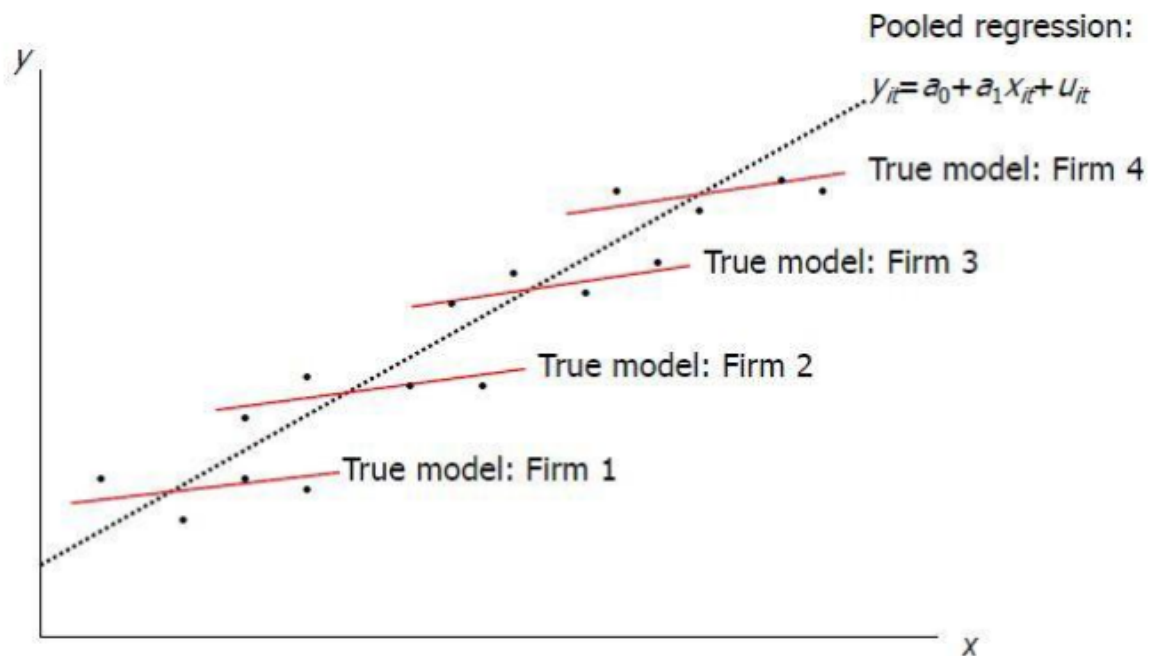


Figure 12: Global Linear Regression Bias

The goal is to completely eliminate unobserved, time-invariant bias (α_i) from your panel data regression so you can get a clean, unbiased estimate of β_1 .

```

library(AER)
data(Fatalities)

# 1. Calculate the missing fatal_rate column first
Fatalities$fatal_rate <- (Fatalities$fatal / Fatalities$pop) * 10000

# 2. Fit a separate fitting parameter for each state (LS )
fatal_fe_lm_mod <- lm(fatal_rate ~ beertax + state - 1, data = Fatalities)
summary(fatal_fe_lm_mod)

```

Call:

```
lm(formula = fatal_rate ~ beertax + state - 1, data = Fatalities)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.58696	-0.08284	-0.00127	0.07955	0.89780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
beertax	-0.65587	0.18785	-3.491	0.000556	***
stateal	3.47763	0.31336	11.098	< 2e-16	***
stateaz	2.90990	0.09254	31.445	< 2e-16	***
statear	2.82268	0.13213	21.364	< 2e-16	***
stateca	1.96816	0.07401	26.594	< 2e-16	***
stateco	1.99335	0.08037	24.802	< 2e-16	***
statect	1.61537	0.08391	19.251	< 2e-16	***
statede	2.17003	0.07746	28.016	< 2e-16	***
statefl	3.20950	0.22151	14.489	< 2e-16	***
statega	4.00223	0.46403	8.625	4.43e-16	***
stateid	2.80861	0.09877	28.437	< 2e-16	***
stateil	1.51601	0.07848	19.318	< 2e-16	***
statein	2.01609	0.08867	22.736	< 2e-16	***
stateia	1.93370	0.10222	18.918	< 2e-16	***
stateks	2.25441	0.10863	20.753	< 2e-16	***
stateky	2.26011	0.08046	28.089	< 2e-16	***
statela	2.63051	0.16266	16.171	< 2e-16	***
stateme	2.36968	0.16006	14.805	< 2e-16	***
statemd	1.77119	0.08246	21.480	< 2e-16	***
statema	1.36788	0.08648	15.818	< 2e-16	***
statemi	1.99310	0.11663	17.089	< 2e-16	***
statemn	1.58042	0.09363	16.880	< 2e-16	***

statems	3.44855	0.20936	16.472	< 2e-16	***
statemo	2.18137	0.09252	23.576	< 2e-16	***
statemt	3.11724	0.09441	33.017	< 2e-16	***
statene	1.95545	0.10551	18.534	< 2e-16	***
statenv	2.87686	0.08106	35.492	< 2e-16	***
statenh	2.22318	0.14114	15.751	< 2e-16	***
statenj	1.37188	0.07333	18.709	< 2e-16	***
statenm	3.90401	0.10154	38.449	< 2e-16	***
stateny	1.29096	0.07563	17.070	< 2e-16	***
statenc	3.18717	0.25173	12.661	< 2e-16	***
statend	1.85419	0.10193	18.191	< 2e-16	***
stateoh	1.80321	0.10193	17.691	< 2e-16	***
stateok	2.93257	0.18428	15.913	< 2e-16	***
stateor	2.30963	0.08117	28.453	< 2e-16	***
statepa	1.71016	0.08648	19.776	< 2e-16	***
stateri	1.21258	0.07753	15.640	< 2e-16	***
statesc	4.03480	0.35479	11.372	< 2e-16	***
statesd	2.47391	0.14121	17.519	< 2e-16	***
statetn	2.60197	0.09162	28.398	< 2e-16	***
statetx	2.56016	0.10853	23.589	< 2e-16	***
stateut	2.31368	0.15453	14.972	< 2e-16	***
statevt	2.51159	0.13973	17.975	< 2e-16	***
stateva	2.18745	0.14664	14.917	< 2e-16	***
statewa	1.81811	0.08233	22.084	< 2e-16	***
statewv	2.58088	0.10767	23.971	< 2e-16	***
statewi	1.71836	0.07746	22.185	< 2e-16	***
statewy	3.24913	0.07233	44.922	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1899 on 287 degrees of freedom

Multiple R-squared: 0.9931, Adjusted R-squared: 0.992

F-statistic: 847.8 on 49 and 287 DF, p-value: < 2.2e-16

```
# 3. Subtract time averaged panel data (Within / Time-Demeaned Transformation)
Fatalities_demeaned <- with(Fatalities,
  data.frame(
    # state is needed to keep track of entities, though not in the final OLS
    state = state,
    fatal_rate = fatal_rate - ave(fatal_rate, state),
    beertax = beertax - ave(beertax, state)
  )
)
```



```
)

# 4. Run the regression on demeaned data without an intercept
summary(lm(fatal_rate ~ beertax - 1, data = Fatalities_demeaned))
```

Call:

```
lm(formula = fatal_rate ~ beertax - 1, data = Fatalities_demeaned)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.58696	-0.08284	-0.00127	0.07955	0.89780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
beertax	-0.6559	0.1739	-3.772	0.000191 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1757 on 335 degrees of freedom

Multiple R-squared: 0.04074, Adjusted R-squared: 0.03788

F-statistic: 14.23 on 1 and 335 DF, p-value: 0.0001913

Week-11-Forecasting

Discrete Response

Binary Response

A binary response (or dichotomous variable) is a discrete response with exactly two possible outcomes, typically coded as 0 (Failure/No) and 1 (Success/Yes).

Integer Response

An integer response (often called count data) is a discrete response where the outcomes are non-negative whole numbers ($\mathbb{N}_0$). Unlike binary data, it can go up to infinity, but it cannot contain decimals

Linear Probability Model

A **Linear Probability Model (LPM)** is a standard Multiple Linear Regression model (Ordinary Least Squares - OLS) applied to a **binary dependent variable** ($Y_i = 0$ or 1). Instead of predicting a continuous outcome, the LPM models the *probability* that an event occurs.

1. Core Specification

The model is written as:

$$Y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \dots + \epsilon_t$$

Where: * $Y_t \in \{0, 1\}$ (e.g., $1 = \text{Recession}$, $0 = \text{No Recession}$) * $X_{k,t}$ are the explanatory variables (which can include time-series elements like lagged variables Y_{t-1}). * The predicted value $\hat{Y}_t$ is interpreted directly as the conditional probability:

$$P(Y_t = 1 \mid X_t) = \hat{Y}_t$$

2. Advantages

- **Simple Interpretation:** Coefficients (β_k) show the direct change in probability. If $\beta_1 = 0.05$, a one-unit increase in X_1 increases the probability of the event by **5 percentage points**.
- **Computational Ease:** Estimated via standard OLS, avoiding the iterative maximum likelihood estimation required for Logit/Probit.
- **Handles Dynamics Well:** Readily accommodates time-series features like fixed effects, instrumental variables, lag structures (Y_{t-1}), and interaction terms without complex scaling issues.

3. Disadvantages & Key Challenges

Nonsensical Predictions

Because the OLS line is straight and unbounded, the model can predict probabilities greater than 1 or less than 0 (e.g., $\hat{Y}_t = 1.25$ or -0.15) for extreme values of X .

Inherent Heteroskedasticity

Since Y_t can only be 0 or 1, the variance of the error term (ϵ_t) depends on X_t :

$$\text{Var}(\epsilon_t | X_t) = P(X_t)[1 - P(X_t)]$$

This violates the OLS homoskedasticity assumption, meaning standard errors and p -values will be biased.

Non-Normal Residuals

The error terms cannot be normally distributed because they can only take on two distinct values for any given X_t ($\epsilon_t = 1 - X_t\beta$ or $\epsilon_t = -X_t\beta$).

Specific Time-Series Issue: Autocorrelation

In time-series applications, the shocks (ϵ_t) are often correlated over time (ϵ_t correlates with ϵ_{t-1}), compounding the bias in standard errors.

4. Crucial Fixes for Empirical Work

To use an LPM validly (especially in time series), you must fix the standard errors: * **Robust Standard Errors:** Use Huber-White sandwich estimators to correct for heteroskedasticity. * **Newey-West Standard Errors:** Use HAC (Heteroskedasticity and Autocorrelation Consistent) standard errors to correct for both heteroskedasticity and time-series serial correlation.

Interpret the prediction of the regression as the probability of a positive answer.

```
# Load data sets
library(AER)
data(HMDA)

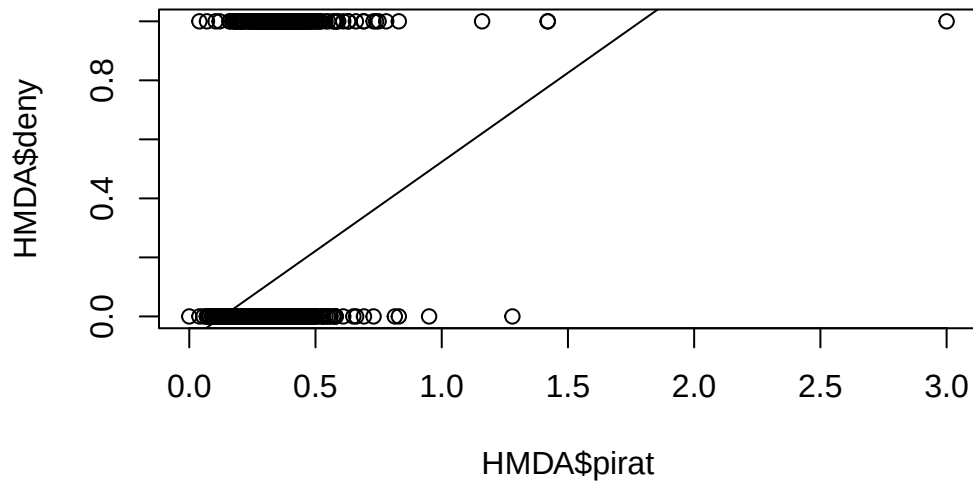
# Convert column 'deny' to numeric
HMDA$deny <- as.numeric(HMDA$deny) - 1

# Estimate a simple linear probability model
linearProbModel <- lm(deny ~ pirat, data = HMDA)
linearProbModel
```

```
Call:
lm(formula = deny ~ pirat, data = HMDA)
```

```
Coefficients:
(Intercept)      pirat
   -0.07991      0.60353
```

```
# Visualize points
plot(x = HMDA$pirat, y = HMDA$deny)
abline(linearProbModel)
```



Is there a Bias?

```
library(AER)
library(lmtest)
data("HMDA")
linearProbModelWithEthnicity <- lm(deny ~ pirat + afam, data = HMDA)
coeftest(linearProbModelWithEthnicity)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.90949	NA	NA	NA
pirat	0.55919	NA	NA	NA
afames	0.17743	NA	NA	NA

According to this model, being Afro-American increases the probability of denial by 17.7%.

Probit Regression

A **Probit Regression** is a statistical model used to estimate the relationship between a set of explanatory variables and a **binary dependent variable** ($Y \in \{0, 1\}$). Unlike the Linear Probability Model (LPM), Probit uses a non-linear S-shaped curve, guaranteeing that all predicted probabilities fall strictly between 0 and 1.

1. Mathematical Specification

The Probit model constrains the predicted values using the **Cumulative Distribution Function (CDF) of the standard normal distribution**, typically denoted by Φ (Phi).

The probability that the event occurs ($Y = 1$) given the predictors X is modeled as:

$$P(Y = 1 | X) = \Phi(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)$$

Where the standard normal CDF is defined by the integral:

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

2. The Latent Variable Framework

The Probit model can be derived from an underlying, unobserved (**latent**) continuous variable Y^* , which represents a propensity or utility index:

$$Y^* = \beta_0 + \beta_1 X_1 + \dots + \epsilon$$

We do not observe Y^* ; we only observe the binary outcome Y , which triggers when Y^* crosses a certain threshold (normalized to 0):

$$Y = \begin{cases} 1 & \text{if } Y^* > 0 \\ 0 & \text{if } Y^* \leq 0 \end{cases}$$

Assuming the error term ϵ follows a **standard normal distribution** ($\epsilon \sim N(0, 1)$) yields the Probit model.

3. Interpretation & Marginal Effects

Because the Probit function is non-linear, the coefficients (β_k) **cannot** be interpreted as direct changes in probability. * A positive β_k means an increase in X_k *increases* the probability that $Y = 1$. * To find the actual change in percentage points, you must calculate **Marginal Effects**:

Method	Definition
Average Marginal Effect (AME)	Calculates the marginal effect for each individual observation and then takes the average across the entire sample. (<i>Preferred approach</i>)
Marginal Effect at the Mean (MEM)	Calculates the marginal effect for a hypothetical individual who holds the exact average characteristics ($\bar{X}$) of the sample.

4. Probit vs. Logit vs. LPM

- **LPM (Linear)**: Easy to interpret but yields nonsensical predictions (< 0 or > 1) and suffers from inherent heteroskedasticity.
- **Probit (Normal CDF)**: Restricts outputs to $[0, 1]$. Assumes normally distributed errors.
- **Logit (Logistic CDF)**: Restricts outputs to $[0, 1]$. Assumes logistically distributed errors (slightly fatter tails than Probit).
- *Note*: In practice, Probit and Logit yield nearly identical conclusions and tracking, varying only in the scale of their raw coefficients.

5. Estimation Method

Unlike OLS used in LPM, Probit coefficients are estimated using **Maximum Likelihood Estimation (MLE)**. MLE searches iteratively for the parameter values (β) that maximize the likelihood of observing the actual data sample.

A probit regression is a statistical model used to estimate the relationship between a set of independent variables and a binary dependent variable, The main goal is to let the linear regression predict the quantile of the error distribution.

```
library(AER)
data(HMDA)
# Estimate with probit model
probitModel<-glm(deny ~ pirat, family = binomial(link = "probit"), data = HMDA)
coeftest(probitModel, vcov. = vcovHC, type = "HC1")
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.19415	0.18901	-11.6087	< 2.2e-16 ***
pirat	2.96787	0.53698	5.5269	3.259e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

How to quantify the impact of the coefficients?

Predict the probability for two different values X and $X + \Delta X$ and compute the difference.

```
# 1. Compute the predictions for P/I ratio = 0.3, 0.4
predictions <- predict(probitModel, newdata = data.frame("pirat" = c(0.3, 0.4)), type = "response")
# 2. Compute the difference in probabilities.
diff(predictions)
```

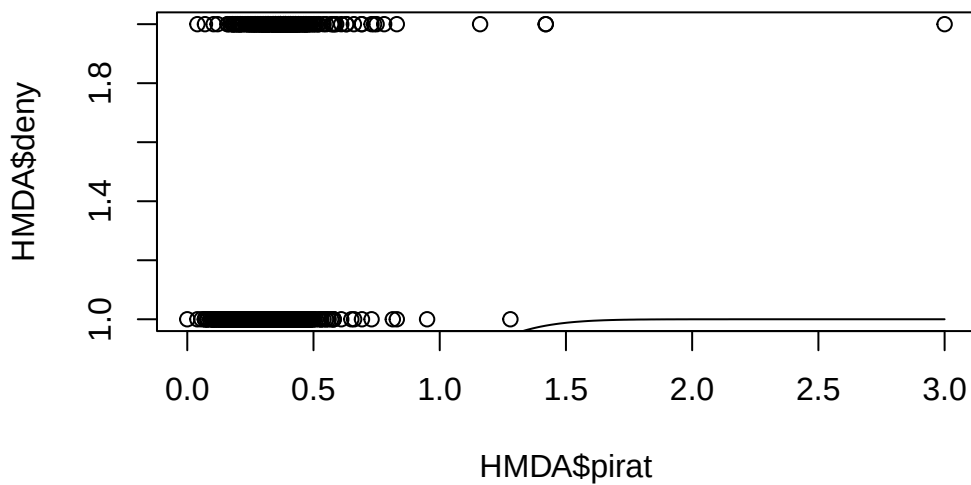
```

      2
0.06081433
```

For payment/income ratio between 0.3 and 0.4, the probability of having the mortgage denied increases by 6.1%.

Visualization of the model

```
# Visualize the probit model
plot(x = HMDA$pirat, y = HMDA$deny)
# Generate predicted values
x<-seq(0, 3, 0.01)
y <- predict(probitModel, list(pirat = x), type = "response")
lines(x, y)
```



Is there an Ethnic Bias?

```
# Ethnicity-aware probit model
probitModelEthn<- glm(deny ~ pirat + afam, family = binomial(link = "probit"), data = HMDA)
coeftest(probitModelEthn)
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.258787	0.136691	-16.5248	< 2.2e-16 ***
pirat	2.741779	0.380469	7.2063	5.749e-13 ***


```
afamyes      0.708155    0.083352    8.4959 < 2.2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 1. Compute the predictions for P/I ratio = 0.3
```

```
predictions <- predict(probitModelEthn, newdata = data.frame("afam" = factor(c("no", "yes")),
```

```
# 2. Compute the difference in probabilities
```

```
diff(predictions)
```

```
2
```

```
0.1578117
```

For a payment/income ratio of 0.3, the probability of having the mortgage denied increases by 15.8 % if the applicant is Afro-American.

Week-12-Discrete-Choice

Logit Regression

A logit regression (short for logistic regression) is a statistical model used to predict the probability of a binary outcome.

Comparing the probit and the logit models

```
# Fitting a logit model
```

```
logitModel <- glm(deny ~ pirat, family = binomial(link = "logit"), data = HMDA)
```

```
coeftest(logitModel, vcov. = vcovHC, type = "HC1")
```

z test of coefficients:

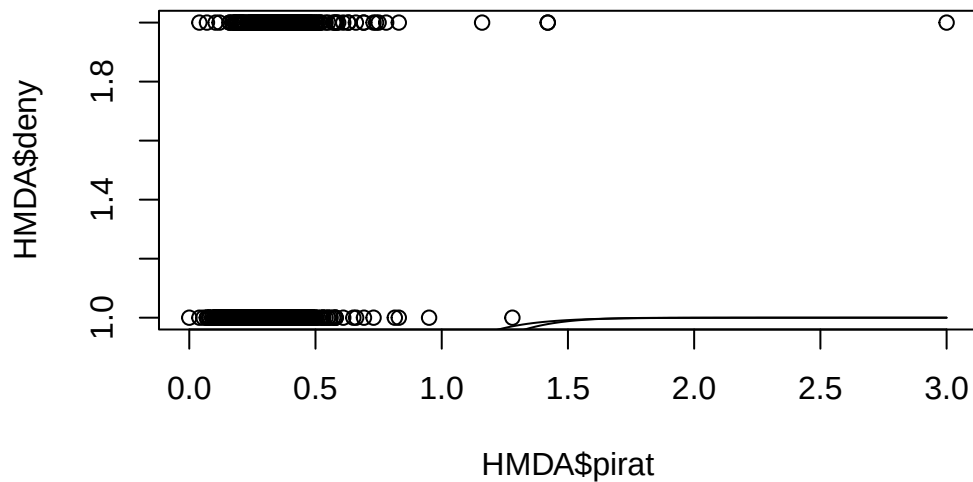
```

              Estimate Std. Error  z value  Pr(>|z|)
(Intercept) -4.02843      0.35898 -11.2218 < 2.2e-16 ***
pirat        5.88450      1.00015   5.8836 4.014e-09 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Visualize the probit and the logit model
plot(x = HMDA$pirat, y = HMDA$deny)

# Generate predicted values
x<-seq(0, 3, 0.01)
y_probit <- predict(probitModel, list(pirat = x), type = "response")
y_logit <- predict(logitModel, list(pirat = x), type = "response")
lines(x, y_probit)
lines(x, y_logit)
```



Is there an ethnic Bias?

```
# Ethnicity-aware logit model
logitModelEthn<- glm(deny ~ pirat + afam, family = binomial(link = "logit"), data = HMDA)

coeftest(logitModelEthn, vcov. = vcovHC, type = "HC1")
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.12556	0.34597	-11.9245	< 2.2e-16 ***
pirat	5.37036	0.96376	5.5723	2.514e-08 ***
afamyes	1.27278	0.14616	8.7081	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# 1. Compute the predictions for P/I ratio = 0.3
predictions <- predict(logitModelEthn,
newdata = data.frame("afam" = factor(c("no", "yes"), levels = levels(HMDA$afam)),
"pirat" = c(0.3, 0.3)),
type = "response")

# 2. Compute the difference in probabilities
diff(predictions)
```

2
0.1492945

For a payment/income ratio of 0.3, the probability having the mortgage denied increases by 14.9% if the applicant is Afro-American (for the probit model it was 15.8%)